

Net Sales Forecasting Case

Implementation Progress Report

Actual Sales, Accrual and CRM Financial Controls Completed

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Organization	Aalto University Executive Education and Professional Development
Solution Platform	Microsoft Power BI Desktop and Power Query
Reporting Period	Financial year 2026
Phase 1 Status	Actual net sales transformation and quality validation completed
Phase 2 Status	Accrual reconciliation, quality and duplicate controls completed
Phase 3 Status	CRM preparation, quality classification, weighted-value reconciliation, opportunity-value reconciliation, current-year weighted-value controls and cross-year allocation validation completed
Current-Year Status Summary	Completed: 85 Match, 39 Not Testable – Missing Current-Year Allocation, and 2 Not Testable – Missing Probability
Allocation Validation	Completed: 126 records classified into 6 allocation statuses with 0 errors and 0 empty classifications
Allocation Status Summary	Completed: 6 grouped statuses reconcile exactly to 126 records
Allocation Exceptions	Completed: 4 non-100% records isolated for targeted review
Missing-Allocation Horizon Analysis	Completed: 52 records reconciled; 41 (78.8%) close in 2026
2026 Missing-Allocation Detail	Completed: 41 near-term records isolated for direct review
2026 Missing-Allocation Exposure	Completed: 41 records represent EUR 4.209M estimated value and EUR 2.318M probability-weighted value
2026 Current-Year Allocation Risk	Completed: 30 records represent EUR 3.572M estimated value and EUR 1.989M probability-weighted value
2026 Forecast Readiness	Completed: 106 opportunities classified; 75 Ready, 28 Missing Current-Year Allocation, 2 Missing Probability and 1 Cross-Year Allocation Review
2026 Forecast Readiness Status Summary	Completed: Ready current-year contribution EUR 807,900.00; one review amount EUR 13,500.00; unresolved weighted pipeline EUR 1,989,178.99
2026 CRM Forecast Bridge	Completed: 4 governed buckets keep EUR 807,900 Ready forecast, EUR 13,500 Review value, EUR 1,989,178.99 weighted risk and EUR 50,000 missing-Probability exposure separate
Business Unit Source Audit	Completed: Actual Sales uses 3 shorter labels; CRM and Accrual share 3 canonical BU labels; 4 Accrual rows remain unassigned

Contents

1	1. Objective	6
2	2. Source Data	11
3	3. Power Query Architecture	11
3.1	3.1 Staging Layer	12
3.2	3.2 Analytical Fact Layer	12
3.3	3.3 Data-Quality Layer	12
3.4	3.4 Dimension Layer	13
3.5	3.5 Semantic-Model Load Scope and Business Unit Relationships	14
3.6	3.6 Shared 2026 Date Dimension	15
4	4. Actual Net Sales Transformation	17
4.1	4.1 Source Structure and Data Types	18
4.2	4.2 Text Standardization	18
4.3	4.3 MonthStart Field	18
5	5. Actual Sales Data-Quality Classification	18
6	6. Actual Project-Month Grain Validation	19
7	7. Actual Sales Analytical Structure and Results	19
7.1	7.1 Standardized Column Names	20
7.2	7.2 Validation Results	20
7.3	7.3 Canonical Business Unit Key Mapping	21
8	8. Ongoing-Project Accrual Transformation	22
8.1	8.1 Source Structure and Data Types	22
8.2	8.2 Text Standardization	22
8.3	8.3 ForecastMonth Field	22
9	9. Accrual Source Total Separation	23
10	10. Accrual Data-Quality Classification	24
11	11. Accrual Analytical Structure	25
11.1	11.1 Canonical Business Unit Key Mapping	26
12	12. Accrual Exact-Duplicate Control	26
12.1	12.1 Duplicate-Control Results	28
13	13. Repeated Accrual Business-Key Analysis	28
13.1	13.1 Candidate Business Key	29
13.2	13.2 Power Query Implementation	29
13.3	13.3 Repeated Business-Key Results	30
13.4	13.4 Detailed Findings	32
13.5	13.5 Repeated-Key Status Classification	32
14	14. CRM Opportunity Preparation and Quality Controls	34
14.1	14.1 Profiling Configuration	34
14.2	14.2 Observed CRM Source Structure	34
14.3	14.3 Profiling Results	35

14.4	14.4 CRM Data-Type Standardization	36
14.5	14.5 CRM Text Standardization and Analytical Renaming	37
14.6	14.6 CRM Opportunity Data-Quality Classification	38
14.7	14.7 CRM Weighted-Value Reconciliation	40
14.8	14.8 Dedicated CRM Weighted-Value Mismatch Control	41
14.9	14.9 Opportunity Value versus Estimated Value Reconciliation	42
14.10	14.10 Current-Year Weighted-Value Reconciliation	43
14.11	14.11 Current-Year Weighted Status Summary Query	47
14.12	14.12 Current-Year and Next-Year Allocation Validation	48
14.13	14.13 Allocation Status Summary Query	50
14.14	14.14 Detailed Allocation Exceptions	51
14.15	14.15 Missing Allocation Horizon Analysis	52
14.16	14.16 Detailed 2026 Missing-Allocation Review	54
14.17	14.17 Financial Exposure of 2026 Missing Allocations	54
14.18	14.18 2026 Current-Year Allocation Risk by Business Unit	55
14.19	14.19 2026 CRM Forecast Readiness by Business Unit	56
14.20	14.20 Forecast Readiness Status Summary	57
14.21	14.21 Governed 2026 CRM Forecast Bridge	58
14.22	14.22 Cross-Source Business Unit Vocabulary Audit	59
14.23	14.23 Canonical CRM Business Unit Key Mapping	60
15	15. Current Implementation Status	61
15.1	15.1 Final Semantic Relationship Audit	62
15.2	15.2 Executive Dashboard End-to-End Business Unit Validation	62
15.3	15.3 Technical Field Visibility Cleanup	63
15.4	15.4 Current Completion Point and Next Control	63
16	16. Governance and Process Value	64
17	17. Next Implementation Phases	66
17.1	17.1 Immediate Weighted-Value Mismatch Investigation	69
17.2	17.2 Outstanding Business Validation	70
17.3	17.3 Completed Current-Year Status Summary	72
17.4	17.4 Completed Allocation Validation	72
17.5	17.5 Completed Allocation Status Summary	73
17.6	17.6 Completed Allocation Exception Isolation	73
17.7	17.7 Completed Missing Allocation Horizon Analysis	74
17.8	17.8 Completed 2026 Missing-Allocation Detail	75
17.9	17.9 Completed 2026 Missing-Allocation Financial Exposure	76
17.10	17.10 Completed 2026 Current-Year Allocation Risk	77
17.11	17.11 Completed 2026 Opportunity Forecast Readiness	78
17.12	17.12 Completed Forecast Readiness Status Summary	80
17.13	17.13 Completed 2026 CRM Forecast Bridge	82
17.14	17.14 Completed Cross-Source Business Unit Vocabulary Audit	84
17.15	17.15 Completed DimBusinessUnit	84
17.16	17.16 Completed Actual Sales Business Unit Key Mapping	85
17.17	17.17 Completed Project Accrual Business Unit Key	86
17.18	17.18 Completed CRM Opportunity Business Unit Key	86
17.19	17.19 Completed Model-Load Cleanup and Business Unit Relationships	87
17.20	17.20 Completed Business Unit Filter-Propagation Validation	87
17.21	17.21 Completed 2026 DimDate Creation	88
17.22	17.22 Completed DimDate Placement and Date Relationships	89

17.23	17.23 Completed Date Filter-Propagation Validation	89
17.24	17.24 Completed Month Sorting and Base Financial Measures	90
17.25	17.25 Completed Business Unit Financial Validation	91
17.26	17.26 Completed Date-Table Designation	92
17.27	17.27 Completed Ready CRM Measures and Forecast Inclusion	92
17.28	17.28 Immediate Next Step – Semantic CRM Governance Measures	94
17.29	17.29 Completed Semantic CRM Governance Measures and KPI Presentation	95
17.30	17.30 Completed Disconnected Forecast-Bucket Dimension	96
17.31	17.31 Completed CRM Calendar Dimension	97
17.32	17.32 Completed CRM Date-Role Relationships	99
17.33	17.33 Immediate Next Step – Validate CRM Date Roles with DAX	100
17.34	17.34 Completed CRM Date-Role DAX Validation Results	101
17.35	17.35 Completed Project Dimension Construction and Profiling	101
17.36	17.36 Immediate Next Step – Govern Project Keys and Relationships	104
17.37	17.37 Completed Project Relationships and Functional Validation	104
17.38	17.38 Immediate Next Step – Project Filter Test and Dashboard Refinement	105
17.39	17.39 Completed Interactive Project Filter Validation	106
17.40	17.40 Completed Top-10 Project Forecast Ranking	107
17.41	17.41 Immediate Next Step – Top-10 Actual versus Accrual Composition	108
17.42	17.42 Completed Top-10 Actual versus Accrual Composition	108
17.43	17.43 Executive Dashboard KPI and Business Unit Control Layer	109
17.44	17.44 Executive Monthly Forecast Visual – Presentation Correction	110
17.45	17.45 Immediate Next Step – Correct Monthly Chart and Add BU Forecast View	110
17.46	17.46 Completed Executive Monthly Forecast Correction	111
17.47	17.47 Completed Executive Business Unit Slicer Validation	112
17.48	17.48 Immediate Next Step – Business Unit Forecast Composition	113
17.49	17.49 Completed Business Unit Forecast Composition	113
17.50	17.50 Completed Forecast Governance Table	114
17.51	17.51 Executive Dashboard Analytical Composition Checkpoint	115
17.52	17.52 Immediate Next Step – Final Interaction Validation	115
17.53	17.53 Completed Business Unit Financial QA	116
17.54	17.54 Completed Final Financial Reconciliation Sign-Off	117
17.55	17.55 Immediate Next Step – Final Presentation Polish and Interview Narrative	118
18	18. Key Interview Message	119

1. Objective

The objective is to establish a controlled, repeatable and auditable foundation for the Aalto EE monthly net sales forecasting solution.

The implementation completed so far includes:

- importing the three provided Excel data sources;
- separating raw staging queries from analytical fact queries;
- preparing and validating the actual net sales dataset;
- creating valid monthly date fields for actuals and accruals;
- implementing row-level data-quality controls;
- validating the project-month grain of actual sales;
- separating the accrual source Total from analytical detail;
- retaining the Total in a dedicated reconciliation query;
- classifying the quality of all 313 accrual detail records;
- validating the accrual detail sum against the source Total;
- identifying and quantifying exact accrual duplicates;
- identifying repeated project-month business keys;
- counting distinct forecast amounts within every repeated key;
- classifying repeated keys into three interpretable patterns;
- profiling all 126 CRM opportunity records across 14 source columns;
- assigning explicit Power Query data types to all 14 CRM fields;
- confirming zero CRM data-type conversion errors;
- standardizing the four CRM text fields using Trim and Clean;
- assigning clear analytical names to all 14 CRM source fields;
- classifying the quality of all 126 CRM opportunity records;
- confirming zero errors and zero empty CRM quality classifications;
- independently recalculating probability-weighted opportunity values;
- reconciling calculated and source weighted values using a EUR 0.01 tolerance;
- identifying one material weighted-value mismatch for investigation;
- reconciling **OpportunityValue** against **EstimatedValue**;
- confirming that the opportunity-value check has one distinct result, **Match**, across all 126 CRM records;
- independently calculating the current-year probability-weighted amount using estimated value, probability and the current-year allocation percentage;

- calculating the source-minus-calculated **CurrentYearWeightedVariance**;
- correcting the Power Query step order so that the variance is created before the final **CurrentYearWeightedCheck**;
- confirming zero errors and zero empty values in **CurrentYearWeightedCheck**; and
- confirming that the current-year check contains three distinct statuses;
- creating **DQ_Opportunity_CurrentYearWeightedStatus** as a dedicated summary query without changing **Fact_Opportunities**;
- confirming the exact current-year status distribution: 85 **Match**, 39 **Not Testable - Missing Current-Year Allocation**, and 2 **Not Testable - Missing Probability**; and
- confirming that all 85 fully testable current-year records reconcile within the EUR 0.01 tolerance, with no **Mismatch** status present;
- creating **DQ_Opportunity_AllocationValidation** as a separate reference query without changing **Fact_Opportunities**;
- calculating **AllocationTotalPct** only when both annual allocation inputs are present;
- classifying all 126 opportunity records into six observed allocation validation statuses with 0 errors and 0 empty classifications;
- confirming that 74 records have both annual allocation percentages present, while 52 records are missing one or both allocation inputs; and
- identifying 70 records with a 100% two-year allocation total, 3 records below 100%, and 1 record above 100% for targeted business review;
- creating **DQ_Opportunity_AllocationStatusSummary** and independently confirming the six allocation-status counts sum to all 126 source records;
- creating **DQ_Opportunity_AllocationExceptions** to isolate the four opportunities whose two-year allocation total is not 100%;
- identifying one 110% over-allocation for **Open custom opportunity 111** in University Programs BU; and
- isolating three below-100% cases with totals of 50%, 50% and 30% for date-aware business interpretation rather than automatic correction;
- creating **DQ_Opportunity_MissingAllocationByYear** and reconciling all 52 missing-allocation records by allocation status, closing year and simulated program-start year;
- confirming that 41 of the 52 missing-allocation records (78.8%) have **ClosingYear = 2026**, so missing allocation is not primarily a distant-horizon issue;
- creating **DQ_Opportunity_2026MissingAllocationDetail** to isolate the 41 near-term records without changing **Fact_Opportunities**; and
- confirming the 2026 closing-year composition as 21 records with both allocations missing, 9 missing only the current-year allocation and 11 missing only the next-year allocation;
- creating **DQ_Opportunity_2026MissingAllocationExposure** to quantify the financial value represented by the 41 near-term allocation gaps;

- confirming EUR 4,208,699.23 of estimated value and EUR 2,317,982.74 of probability-weighted value across those 41 records;
- creating `DQ_Opportunity_2026CurrentYearAllocationRisk` to isolate the 30 2026-closing opportunities whose current-year allocation is missing;
- confirming EUR 3,572,367.98 of estimated value and EUR 1,989,178.99 of probability-weighted value in that current-year allocation risk set; and
- confirming that ExEd Programs BU contains the largest share of this weighted risk, EUR 1,229,613.75 across 18 records;
- creating `DQ_Opportunity_2026ForecastReadinessSummary` to classify all 106 CRM opportunities closing in 2026 by forecast readiness and Business Unit;
- confirming 75 Ready - Reconciled Current-Year Value records, 28 Not Ready - Missing Current-Year Allocation records, 2 Not Ready - Missing Probability records and 1 Review - Cross-Year Allocation Above 100% record;
- confirming that the 75 Ready opportunities contribute EUR 807,900.00 of reconciled current-year CRM value;
- creating `DQ_Opportunity_2026ForecastReadinessStatusSummary` to collapse the Business Unit detail into four auditable forecast-readiness buckets; and
- preserving EUR 1,989,178.99 of unresolved weighted pipeline and EUR 50,000.00 of missing-probability estimated value outside the Ready forecast instead of silently converting them into current-year revenue;
- creating `DQ_Opportunity_2026ForecastBridge` and separating the four forecast treatments into dedicated financial columns;
- confirming the governed bridge values as EUR 807,900.00 Ready current-year forecast, EUR 13,500.00 held for review, EUR 1,989,178.99 at-risk weighted pipeline and EUR 50,000.00 estimated pipeline with missing Probability;
- creating `DQ_BusinessUnit_SourceValues` to compare Business Unit vocabularies across Actual Sales, Project Accrual and CRM Opportunities;
- confirming that Actual Sales uses ExEd, Qualification Programs and University Programs, while CRM and Accrual use the corresponding ... Programs BU labels; and
- confirming that 4 accrual records still have a missing Business Unit and must remain explicitly unassigned until business validation;
- creating the 04_Dimensions query group and the canonical `DimBusinessUnit` dimension with four governed members;
- assigning Business Unit keys 1, 2 and 3 to the three populated canonical Business Units and key 0 to Unassigned / Missing BU;
- adding `BusinessUnitKey` to `Fact_ActualSales` while preserving the original `BusinessUnitSource` field for lineage; and
- confirming that all 795 Actual Sales rows have a valid non-null key: 346 rows map to key 1, 120 rows to key 2 and 329 rows to key 3;
- adding `BusinessUnitKey` to `Fact_ProjectAccrual` while preserving `BusinessUnitSource`;

- confirming that all 313 Accrual rows have valid keys, including the 4 missing-BU rows assigned explicitly to key 0;
- adding **BusinessUnitKey** to **Fact_Opportunities** while preserving the original CRM Business Unit label;
- confirming that all 126 Opportunity rows have valid keys across the three canonical Business Units with no key-0 Opportunity records;
- disabling model load for staging and data-quality queries while retaining their Power Query logic and refresh lineage;
- reducing the visible semantic model to **DimBusinessUnit**, **Fact_ActualSales**, **Fact_ProjectAccrual** and **Fact_Opportunities**; and
- creating three active single-direction Business Unit relationships with the dimension on the one side and each fact on the many side;
- validating Business Unit filter propagation with three DAX row-count measures and confirming exact reconciliation to 795 Actual, 313 Accrual and 126 CRM Opportunity rows;
- confirming the Business Unit split as 346/162/92 for ExEd, 120/43/7 for Qualification Programs, 329/104/27 for University Programs and 0/4/0 for the Unassigned member across Actual/Accrual/CRM respectively; and
- creating **DimDate** for the full 2026 calendar with 365 unique daily dates, 12 months, 12 Year-Month values, 12 MonthStart values and 4 quarters, with zero errors and zero empty dates;
- moving **DimDate** into the governed dimension layer;
- creating active Single-direction Date relationships from **DimDate[Date]** to **Fact_ActualSales[MonthStart]** and **Fact_ProjectAccrual[ForecastMonth]**; and
- validating monthly Date-filter propagation across both facts, confirming 795 Actual rows in January–August and 313 Accrual rows in September–December with no cross-period leakage;
- sorting **DimDate[MonthName]** chronologically by **MonthNumber**;
- creating the DAX measures **Actual Net Sales**, **Project Accrual Forecast** and **Base Full-Year Forecast**; and
- reconciling the completed base full-year forecast to EUR 14,635,067.81 of Actual Net Sales plus EUR 6,616,025.78 of Project Accrual Forecast, producing EUR 21,251,093.59 before CRM contribution;
- formally marking **DimDate** as the model Date table using **DimDate[Date]**;
- validating the three base financial measures by Business Unit and reconciling the BU rows exactly to EUR 21,251,093.59;
- creating **CRM Ready Opportunity Count** and **CRM Ready Current-Year Value** measures and confirming 75 Ready opportunities with EUR 807,900.00 of current-year value; and
- creating **Forecast Including Ready CRM** and reconciling the current governed annual forecast to EUR 22,058,993.59;
- creating and validating the semantic CRM governance measures for EUR 13,500.00 of Review value, EUR 1,989,178.99 of At-Risk weighted pipeline and EUR 50,000.00 of Missing-Probability

estimated exposure;

- creating the optional **Forecast If Review Approved** scenario and reconciling it to EUR 22,072,493.59;
- creating the disconnected **DimForecastBucket** presentation dimension to expose governed forecast treatments without adding an ambiguous relationship path;
- creating **DimCRMDate** for the complete 2024–2028 CRM horizon with 1,827 unique daily dates and zero errors;
- creating one active CRM Date relationship on **ClosingDate** and one inactive role-playing relationship on **SimulatedProgramStartDate**, both using Single cross-filter direction;
- validating both CRM Date roles with DAX and confirming that each role reconciles to all 126 opportunity records while producing different yearly distributions;
- creating **DimProject** from Actual Sales and Project Accrual ProjectIDs; and
- confirming that **DimProject** contains 252 rows, 252 distinct and 252 unique ProjectIDs, with zero errors and zero empty values;
- moving **DimProject** into the governed 04_Dimensions query group with model load retained;
- creating two active Single-direction Project relationships directly on the clean natural key **ProjectID**, with **DimProject** on the one side and Actual Sales / Project Accrual on the many side;
- validating Project-level filter propagation and reconciling the relationship layer to 795 Actual rows, 313 Accrual rows, EUR 14,635,067.81 of Actual Net Sales, EUR 6,616,025.78 of Project Accrual Forecast and EUR 21,251,093.59 of Base Full-Year Forecast; and
- retaining **ProjectKey** in **DimProject** as a technical surrogate identifier while deliberately avoiding an unnecessary fact-table merge because the supplied fact **ProjectID** values already reconcile cleanly to the unique dimension key.
- completing the final interactive Project slicer validation with 07901009, confirming EUR 439,132.89 Actual Net Sales, EUR 308,913.00 Project Accrual Forecast and EUR 748,045.89 Base Full-Year Forecast;
- confirming that the Project slicer does not create an unsupported CRM-to-Project filter path and that CRM forecast-governance measures remain independent; and
- creating the Top-10 Project ranking chart using **DimProject[ProjectID]** and **Base Full-Year Forecast**, sorted from highest to lowest forecast value.
- completing the companion Top-10 Project composition chart using the same governed Project dimension and the existing **Actual Net Sales** and **Project Accrual Forecast** measures;
- creating a dedicated **Executive Dashboard** report page with eight EUR-formatted forecast and CRM-governance KPI cards;
- adding a governed Business Unit slicer based on **DimBusinessUnit[BusinessUnitName]**; and
- adding the 12-month Actual-versus-Accrual forecast visual and identifying the remaining presentation correction: replace the current 100% stacked column chart with a standard stacked column chart so the y-axis represents financial magnitude rather than percentage composition.

- completing that presentation correction by changing the monthly visual to a standard stacked column chart with a monetary EUR axis;
- interactively validating the Executive Dashboard with each governed Business Unit member and confirming consistent filtering across Actual Sales, Project Accrual and CRM governance measures; and
- confirming that the explicit **Unassigned / Missing** BU member isolates EUR 42,550.00 of Accrual value without fabricating Actual or CRM value.
- completing the Business Unit forecast-composition chart using the governed **DimBusinessUnit** dimension and the validated **Actual Net Sales** and **Project Accrual Forecast** measures;
- adding the **Forecast Governance - Included vs At Risk** table with five explicit governed forecast buckets and their treatment descriptions; and
- bringing the Executive Dashboard to a near-final analytical composition containing annual KPIs, organizational composition, monthly timing, Project concentration and forecast-governance context.
- completing a final semantic-model relationship audit covering all nine relationships and confirming eight active relationships plus the intentionally inactive CRM program-start date role;
- completing an end-to-end Business Unit slicer validation across the newly added Governance, Business Unit and Top-10 Project visuals for all four governed Business Unit members; and
- hiding technical keys, sort helpers and selected data-quality diagnostics from the business-facing field list while preserving them in the model for audit and maintenance.

2. Source Data

The following source datasets were imported into Power BI:

Dataset	Purpose
Net Sales Actuals	Actual net sales by Business Unit, project and accounting month.
Accruals	Forecasted accrued net sales for ongoing projects.
Open Custom Opportunities	Open CRM sales opportunities, including estimated value, probability and expected annual allocation.

Inference – Source Scope

The three datasets represent complementary forecast layers rather than interchangeable inputs. Actual sales provide recognized revenue, accruals provide the remaining forecast for ongoing projects, and CRM opportunities provide probability-adjusted future potential. They must therefore be validated independently before being combined in one forecast model.

3. Power Query Architecture

A staging-and-fact architecture was implemented to separate source ingestion from analytical transformation and validation.

3.1 Staging Layer

The raw staging queries are:

- stg_Actuals_Raw
- stg_Accruals_Raw
- stg_Opportunities_Raw

Loading is disabled for these staging queries. They remain part of the refresh process and preserve traceability to the original Excel sources.

3.2 Analytical Fact Layer

Reference queries were created from the staging layer:

- Fact_ActualSales
- Fact_ProjectAccrual
- Fact_Opportunities

For example, the actual-sales and accrual fact queries reference their corresponding staging queries:

```
= stg_Actuals_Raw
```

```
= stg_Accruals_Raw
```

3.3 Data-Quality Layer

Dedicated control queries were created to keep validation logic separate from the analytical facts:

- DQ_Actual_DuplicateKeys;
- DQ_Accrual_SourceTotal;
- DQ_Accrual_ExactDuplicates;
- DQ_Accrual_RepeatedBusinessKeys;
- DQ_Opportunity_WeightedValueMismatch;
- DQ_Opportunity_CurrentYearWeightedStatus;
- DQ_Opportunity_AllocationValidation;
- DQ_Opportunity_AllocationStatusSummary;
- DQ_Opportunity_AllocationExceptions;
- DQ_Opportunity_MissingAllocationByYear;
- DQ_Opportunity_2026MissingAllocationDetail;
- DQ_Opportunity_2026MissingAllocationExposure;
- DQ_Opportunity_2026CurrentYearAllocationRisk;
- DQ_Opportunity_2026ForecastReadinessSummary;
- DQ_Opportunity_2026ForecastReadinessStatusSummary;
- DQ_Opportunity_2026ForecastBridge; and

- DQ_BusinessUnit_SourceValues.

Loading has now been disabled for the data-quality queries before semantic-model construction. The queries remain available in Power Query with all applied steps intact, but they no longer appear as report-model tables.

This layered architecture provides:

- transparent source-to-report lineage;
- separation between raw, analytical and control data;
- repeatable refresh operations;
- auditable transformation steps;
- easier future replacement of source systems; and
- cleaner tables in the semantic model.

Inference – Query Architecture

The separation of staging, analytical facts and data-quality controls creates a traceable refresh path. Raw source evidence remains unchanged while business transformations and exception logic can evolve without obscuring lineage.

3.4 Dimension Layer

A dedicated dimension group has now been added:

04_Dimensions

The first shared dimension is:

DimBusinessUnit

It contains three canonical Business Units and one explicit technical member for records whose Business Unit is missing. The dimension is intentionally separate from the facts so source labels can be preserved while relationships use stable numeric keys.

A second shared dimension has now been created:

DimDate

It covers every day from 1 January 2026 through 31 December 2026 and provides daily, monthly and quarterly attributes for common time filtering. The query has now been positioned in the governed dimension layer before Date relationships were finalized.

A separate CRM calendar has subsequently been added:

DimCRMDate

It covers the complete 2024–2028 opportunity horizon and supports explicit role-playing analysis of CRM closing dates and simulated program-start dates.

A disconnected presentation dimension has also been added:

DimForecastBucket

It is intentionally not related to the facts and is used only to present the governed forecast bridge and scenario-treatment buckets.

The Project dimension has now been constructed in Power Query as:

```
DimProject
```

Its profile contains 252 ProjectIDs, all distinct and unique, with zero errors and zero empty values. The query has now been moved into 04_Dimensions and retained as a loaded semantic-model dimension. Although the table also contains a technical ProjectKey, the final relationships use the clean natural key ProjectID directly because that field is already present in both project-based facts and is unique on the Dimension side.

Inference – Dimension Layer

Introducing a shared Business Unit dimension removes dependence on source- specific text labels. Actual Sales can retain its shorter source vocabulary, CRM and Accrual can retain their existing labels, and all three facts can still filter consistently through the same governed Business Unit keys.

3.5 Semantic-Model Load Scope and Business Unit Relationships

Before relationship construction, model load was disabled for the staging and data-quality queries. Their transformation and validation logic remains in Power Query. At that stage the visible semantic model was reduced to DimBusinessUnit plus the three analytical facts. After the shared Date dimension was created and loaded, the current visible model contains:

- DimBusinessUnit;
- DimDate;
- Fact_ActualSales;
- Fact_ProjectAccrual; and
- Fact_Opportunities.

Subsequent modelling phases added DimCRMDate, the disconnected DimForecastBucket presentation table and DimProject. The current core star-schema dimensions therefore include Business Unit, 2026 Date, CRM Date roles and Project, while DimForecastBucket remains intentionally disconnected.

Three Business Unit relationships were then created:

Dimension Side	Fact Side	Cardinality	Filter
DimBusinessUnit[BusinessUnitKey]	Fact_ActualSales[BusinessUnitKey]	*	Single
DimBusinessUnit[BusinessUnitKey]	Fact_ProjectAccrual[BusinessUnitKey]	*	Single
DimBusinessUnit[BusinessUnitKey]	Fact_Opportunities[BusinessUnitKey]	*	Single

All three relationship lines are active. In the Power BI relationship editor, the same structure may be displayed from the fact-table perspective as Many to one (*:1), with DimBusinessUnit on the one side. The selected Actual Sales relationship is explicitly confirmed as active with Cross-filter direction = Single.

Inference – Business Unit Semantic Model

The Business Unit portion of the model now follows a controlled star-schema pattern: one governed dimension filters three separate facts, while fact-to-fact relationships are avoided. Disabling load for staging and DQ queries removes technical tables from the report model without deleting their audit logic. Single-direction relationships reduce ambiguity and establish a clean basis for shared Business Unit slicing across actuals, accruals and CRM opportunities.

3.6 Shared 2026 Date Dimension

A daily Date dimension was created to provide one governed calendar for Actual Sales and Project Accrual reporting:

```
let
  StartDate = #date(2026, 1, 1),
  EndDate = #date(2026, 12, 31),

  DateList =
    List.Dates(
      StartDate,
      Duration.Days(EndDate - StartDate) + 1,
      #duration(1, 0, 0, 0)
    ),

  ToTable =
    Table.FromList(
      DateList,
      Splitter.SplitByNothing(),
      {"Date"}
    ),

  ChangedType =
    Table.TransformColumnTypes(
      ToTable,
      {
        {"Date", type date}
      }
    ),

  AddedYear =
    Table.AddColumn(
      ChangedType,
      "Year",
      each Date.Year([Date]),
      Int64.Type
    ),

  AddedMonthNumber =
    Table.AddColumn(
      AddedYear,
      "MonthNumber",
      each Date.Month([Date]),
      Int64.Type
    ),

  AddedMonthName =
    Table.AddColumn(
      AddedMonthNumber,
```

```
        "MonthName",
        each Date.MonthName([Date]),
        type text
    ),

    AddedYearMonth =
        Table.AddColumn(
            AddedMonthName,
            "YearMonth",
            each Date.Year([Date]) * 100 + Date.Month([Date]),
            Int64.Type
        ),

    AddedMonthStart =
        Table.AddColumn(
            AddedYearMonth,
            "MonthStart",
            each Date.StartOfMonth([Date]),
            type date
        ),

    AddedQuarter =
        Table.AddColumn(
            AddedMonthStart,
            "Quarter",
            each "Q" & Text.From(Date.QuarterOfYear([Date])),
            type text
        )

in
    AddedQuarter
```

The completed profile confirms:

Date-Dimension Metric	Result
Calendar period	1 January 2026 – 31 December 2026
Rows	365
Date valid values	100%
Date errors	0
Date empty values	0
Distinct / unique Date values	365 / 365
Distinct Year values	1
Distinct MonthNumber / MonthName values	12 / 12
Distinct YearMonth values	12
Distinct MonthStart values	12
Distinct Quarter values	4

Date-Dimension Relationship Result

`DimDate` is now used as a governed semantic-model dimension. Two active Date relationships connect the calendar to Actual Sales and Project Accrual:

- `Fact_ActualSales[MonthStart]` many-to-one `DimDate[Date]`; and
- `Fact_ProjectAccrual[ForecastMonth]` many-to-one `DimDate[Date]`.

Both relationships use `Cross-filter direction = Single` and are active.

Inference – Shared Date Dimension

The 2026 calendar is complete, unique and error-free and now functions as the shared time dimension for the two monthly financial facts. Actual Sales and Project Accrual remain separate facts, but both can be filtered through one calendar. CRM Opportunity dates are intentionally not connected at this stage because the CRM table contains multiple date roles and records outside 2026; forcing one active CRM Date relationship now would create an unnecessary modelling constraint.

4. Actual Net Sales Transformation

The `Fact_ActualSales` query was prepared using a controlled transformation sequence.

4.1 Source Structure and Data Types

Source Column	Data Type	Business Meaning
BU	Text	Source Business Unit label
Cost center (prj)	Text	Project-related cost center
Project	Text	Project identifier
Year-Month	Whole Number	Accounting year and month
Net sales	Decimal Number	Actual recognized net sales

4.2 Text Standardization

Trim and Clean transformations were applied to the Business Unit, cost center and project columns. This removes leading or trailing spaces and non-printing characters without changing valid business values.

The corresponding Power Query steps were documented as:

```
Trimmed Text  
Cleaned Text
```

4.3 MonthStart Field

The numeric Year-Month field was converted into a valid month-level date:

```
let  
    ym = Text.PadStart(Text.From(["Year-Month"]), 6, "0")  
in  
    #date(  
        Number.FromText(Text.Start(ym, 4)),  
        Number.FromText(Text.End(ym, 2)),  
        1  
    )
```

The resulting column was named **MonthStart** and its data type was set to **Date**. The actual-sales period covers January through August 2026.

Inference – Actual-Sales Transformation

The actual-sales source has been converted into a consistent monthly fact structure. The valid **MonthStart** field enables reliable time filtering, date relationships and comparison with the September–December forecast period.

5. Actual Sales Data-Quality Classification

A row-level quality classification was implemented to distinguish critical data problems from financially meaningful values requiring review.

```
if [BU] = null or Text.Trim(Text.From([BU])) = "" then  
    "Critical - Missing BU"  
else if [Project] = null  
    or Text.Trim(Text.From([Project])) = "" then  
    "Critical - Missing Project"  
else if [MonthStart] = null then  
    "Critical - Invalid Month"  
else if ["Net sales"] = null then  
    "Critical - Missing Net Sales"
```

```
else if [#Net sales] < 0 then
    "Review - Negative Amount"
else if [#Net sales] = 0 then
    "Review - Zero Amount"
else
    "Valid"
```

The resulting column was named **ActualQualityStatus**.

Zero and negative net sales values were not automatically removed. Negative values may represent credit notes, accounting corrections or revenue reversals. These records were retained and classified as review items for Finance validation.

Inference – Actual-Sales Quality

The absence of critical issues means all 795 actual-sales records contain the minimum identifiers and values required for analysis. The 118 zero or negative records are accounting review items rather than technical failures and should remain visible until their financial meaning is confirmed.

6. Actual Project-Month Grain Validation

The intended analytical grain of the actual-sales fact is:

One row per project per accounting month

A composite project-month key was created:

```
if [Project] = null or [MonthStart] = null then
    null
else
    Text.Upper(Text.Trim(Text.From([Project])))
    & "|"
    & Date.ToText([MonthStart], "yyyyMM")
```

The resulting field was named **ProjectMonthKey**.

The dedicated duplicate-control query grouped the records by this field. Only keys with a record count greater than one were retained. The query was named **DQ_Actual_DuplicateKeys**.

The result was:

0 duplicate project-month keys

This confirms that the actual-sales table currently complies with the intended grain.

Inference – Actual-Sales Grain

No project-month duplication was detected. Consequently, actual net sales can be aggregated by project and month without a duplicate-driven overstatement at the validated source grain.

7. Actual Sales Analytical Structure and Results

7.1 Standardized Column Names

Source Name	Analytical Name
BU	BusinessUnitSource
Cost center (prj)	CostCenter
Project	ProjectID
Year-Month	YearMonth
Net sales	ActualNetSales

The final Fact_ActualSales structure now contains:

- BusinessUnitSource;
- CostCenter;
- ProjectID;
- YearMonth;
- ActualNetSales;
- MonthStart;
- ActualQualityStatus;
- ProjectMonthKey; and
- BusinessUnitKey.

7.2 Validation Results

Validation Metric	Result
Total source records	795
Valid records	677
Zero-value review records	102
Negative-value review records	16
Critical data-quality issues	0
Missing quality classifications	0
Duplicate project-month keys	0
Available accounting period	January–August 2026
Total actual net sales	EUR 14,635,067.81

Actual Sales Result

The actual net sales dataset is structurally valid, traceable and ready for integration into the forecasting semantic model.

All 795 records were retained. Zero and negative accounting values remain visible for Finance review.

Inference – Actual-Sales Readiness

The actual-sales fact is structurally ready for the semantic model. Its total of EUR 14,635,067.81 is supported by complete row-level classifications and a validated project-month key, providing a dependable year-to-date baseline for the forecast.

7.3 Canonical Business Unit Key Mapping

After DimBusinessUnit was created, a numeric BusinessUnitKey was added to Fact_ActualSales. The original BusinessUnitSource text field was retained for lineage.

The implemented mapping is:

```
if [BusinessUnitSource] = "ExEd" then
    1
else if [BusinessUnitSource] = "Qualification Programs" then
    2
else if [BusinessUnitSource] = "University Programs" then
    3
else
    0
```

The new field was explicitly typed as Whole Number / Int64.Type. The completed profile shows:

Actual Sales BU-Key Metric	Result
Total Actual Sales rows	795
BusinessUnitKey valid values	100%
BusinessUnitKey errors	0
BusinessUnitKey empty values	0
Distinct BusinessUnitKey values	3
Key 1 – ExEd Programs BU	346 rows
Key 2 – Qualification Programs BU	120 rows
Key 3 – University Programs BU	329 rows
Key 0 – Unassigned / Missing BU	0 rows

Inference – Actual Sales Business Unit Mapping

All 795 Actual Sales records now have a stable numeric Business Unit key while the source label remains available for traceability. The mapping reproduces the earlier source-value audit exactly: 346 ExEd rows, 120 Qualification Programs rows and 329 University Programs rows. No Actual Sales record falls into the Unassigned member, so the first fact is ready for a one-to-many relationship from DimBusinessUnit.

8. Ongoing-Project Accrual Transformation

The second implementation phase prepared the ongoing-project accrual forecast contained in Fact_ProjectAccrual.

8.1 Source Structure and Data Types

Source Column	Data Type	Business Meaning
BU	Text	Source Business Unit label
Accrual Basis	Text	Basis or rule used for the project accrual
PNo.	Text	Project number
Net Sales Year	Whole Number	Forecast financial year
Net Sales Month	Whole Number	Forecast financial month
Accrued Net Sales Forecast	Decimal Number	Forecasted net sales for the project-month

8.2 Text Standardization

Trim and Clean transformations were applied to BU, Accrual Basis and PNo.. This prevents false differences caused by whitespace or non-printing characters.

8.3 ForecastMonth Field

The forecast year and month were combined into a valid monthly date:

```
try
    #date(
        Int64.From(["Net Sales Year"]),
        Int64.From(["Net Sales Month"]),
        1
    )
otherwise
    null
```

The resulting column was named ForecastMonth and its data type was set to Date. The source forecast period covers September through December 2026.

Inference – Accrual Transformation

The accrual source now has a reliable monthly date at the same calendar level as actual sales. Its September–December coverage complements the January–August actual-sales period and supports a full-year 2026 forecast when the two fact layers are combined through a shared Date dimension.

9. Accrual Source Total Separation

The accrual source contains a Total row. Including this row in the analytical fact would double-count the forecast. A Boolean helper field was therefore created:

```
if [BU] = null then
  false
else
  Text.Upper(Text.Trim(Text.From([BU]))) = "TOTAL"
```

The resulting field was named IsTotalRow.

The source Total was retained in the dedicated control query:

```
DQ_Accrual_SourceTotal
```

This control query contains only records where:

```
IsTotalRow = true
```

The analytical fact was filtered using:

```
IsTotalRow = false
```

The validation results are:

Accrual Control Metric	Result
Original source records	314
Source Total rows identified	1
Detail records retained in fact	313
Source control total	EUR 6,616,025.78
Sum of all 313 detail records	EUR 6,616,025.78
Reconciliation variance	EUR 0.00
Classification errors	0
Empty classifications	0
Forecast period	September–December 2026

Total-Row Control Result

The source Total has been separated from the analytical detail. It remains available for reconciliation while only the 313 project-month detail records are retained in Fact_ProjectAccrual. The helper field IsTotalRow can now be removed from the final analytical fact structure.

Inference – Source-Total Reconciliation

The detail sum reconciles exactly to the source control total, with a variance of EUR 0.00. This proves that filtering the Total row prevents double-counting without losing financial value or breaking source-to-report reconciliation.

10. Accrual Data-Quality Classification

A row-level data-quality field was created in Fact_ProjectAccrual:

```
if [BU] = null
    or Text.Trim(Text.From([BU])) = "" then
    "Critical - Missing BU"
else if [#PNo.] = null
    or Text.Trim(Text.From([#PNo.])) = "" then
    "Critical - Missing Project"
else if [ForecastMonth] = null then
    "Critical - Invalid Forecast Month"
else if [#Accrued Net Sales Forecast] = null then
    "Critical - Missing Forecast"
else if [#Accrued Net Sales Forecast] < 0 then
    "Review - Negative Forecast"
else if [#Accrued Net Sales Forecast] = 0 then
    "Review - Zero Forecast"
else
    "Valid"
```

The resulting column was named AccrualQualityStatus.

Accrual Quality Status	Record Count
Valid	308
Critical – Missing BU	4
Review – Zero Forecast	1
Critical – Missing Project	0
Critical – Invalid Forecast Month	0
Critical – Missing Forecast	0
Review – Negative Forecast	0
Total detail records	313

Four accrual records have no Business Unit classification. Their combined forecast value is EUR 42,550.00.

These records were not removed or assigned automatically. They remain visible as controlled exceptions and require validation against a project master or confirmation from Finance.

One zero-value forecast was also retained because it may represent a valid project-month with no expected revenue.

Inference – Accrual Quality

Most accrual records are analytically usable, but Business Unit completeness is not yet perfect. The four unclassified records represent EUR 42,550.00 and must remain visible as controlled exceptions until they can be matched to a project master or confirmed by Finance.

11. Accrual Analytical Structure

The following standardized analytical names are used in the accrual fact table:

Source Name	Analytical Name
BU	BusinessUnitSource
Accrual Basis	AccrualBasis
PNo.	ProjectID
Net Sales Year	ForecastYear
Net Sales Month	ForecastMonthNumber
Accrued Net Sales Forecast	ProjectAccrualForecast
ForecastMonth	ForecastMonth
AccrualQualityStatus	AccrualQualityStatus

After renaming the source fields and removing the temporary `IsTotalRow` helper, the final `Fact_ProjectAccrual` structure now contains:

- `BusinessUnitSource`;
- `AccrualBasis`;
- `ProjectID`;
- `ForecastYear`;
- `ForecastMonthNumber`;
- `ProjectAccrualForecast`;
- `ForecastMonth`;
- `AccrualQualityStatus`; and
- `BusinessUnitKey`.

Inference – Accrual Analytical Structure

The standardized fields provide clear semantic names and a stable grain for modelling. Source-specific labels remain available for lineage, while the canonical Business Unit key supports controlled dimensional filtering.

11.1 Canonical Business Unit Key Mapping

A numeric `BusinessUnitKey` was added to `Fact_ProjectAccrual` using the canonical Business Unit vocabulary:

```
if [BusinessUnitSource] = null
    or Text.Trim(Text.From([BusinessUnitSource])) = "" then
    0
else if [BusinessUnitSource] = "ExEd Programs BU" then
    1
else if [BusinessUnitSource] = "Qualification Programs BU" then
    2
else if [BusinessUnitSource] = "University Programs BU" then
    3
else
    null
```

The field was typed as Whole Number / `Int64.Type`. The completed column profile contains 313 rows, 100% valid values, 0 errors, 0 empty values and 4 distinct keys.

The source-value audit reconciles the key distribution to:

- key 1: 162 ExEd Programs BU records;
- key 2: 43 Qualification Programs BU records;
- key 3: 104 University Programs BU records; and
- key 0: 4 records whose Business Unit is missing.

Inference – Project Accrual Business Unit Mapping

All 313 accrual records can now participate in dimensional Business Unit filtering without fabricating a classification for incomplete source rows. The four missing-BU records are deliberately isolated under key 0, preserving their EUR 42,550.00 forecast exposure as an explicit unassigned exception rather than hiding it inside a real Business Unit.

12. Accrual Exact-Duplicate Control

A dedicated data-quality query was created from `Fact_ProjectAccrual`:

```
DQ_Accrual_ExactDuplicates
```

The query grouped the accrual records using all six fields that define an exact duplicate:

- `BusinessUnitSource`;
- `AccrualBasis`;
- `ProjectID`;
- `ForecastYear`;

- ForecastMonthNumber; and
- ProjectAccrualForecast.

The grouped records were counted in `RecordCount`. Only groups with more than one source record were retained:

```
= Table.SelectRows(  
    #"Grouped Rows",  
    each [RecordCount] > 1  
)
```

Two additional diagnostic fields were created:

```
DuplicateExcessCount = [RecordCount] - 1
```

```
PotentialOverstatement =  
    [DuplicateExcessCount] * [ProjectAccrualForecast]
```

12.1 Duplicate-Control Results

The completed control produced the following results:

Duplicate-Control Metric	Result
Exact duplicate groups	8
Source records involved	20
Excess duplicate records	12
Minimum records per duplicate group	2
Maximum records per duplicate group	3
Projects represented	2
Business Unit represented	University Programs BU
Accrual basis represented	Actual Close Date
Affected forecast period	September–December 2026
Potential duplicate overstatement	EUR 175,318.63

Four duplicate groups relate to project L9109051. Each monthly record appears three times. The other four groups relate to project L9109067. The exact duplicated amount appears twice in each affected month.

The potential overstatement of EUR 175,318.63 is a diagnostic amount. It must not be treated as a confirmed correction before Finance validates the business meaning of the repeated source records.

No records were deleted automatically from **Fact_ProjectAccrual**. The source-aligned amount remains available for reconciliation and transparent review.

Inference – Exact-Duplicate Control

The control identifies a concentrated duplication risk: 8 exact groups, 20 involved rows and 12 excess rows across only two projects. The EUR 175,318.63 potential overstatement is a risk estimate, not an approved correction, because Finance must confirm the business meaning before any source-aligned value is removed.

13. Repeated Accrual Business-Key Analysis

A second accrual control was implemented to distinguish repeated project-month business keys from exact duplicate records. The dedicated control query was named:

```
DQ_Accrual_RepeatedBusinessKeys
```

13.1 Candidate Business Key

The accrual records were grouped using the following candidate business key:

- BusinessUnitSource;
- AccrualBasis;
- ProjectID;
- ForecastYear; and
- ForecastMonthNumber.

The forecast amount was intentionally excluded from the grouping key. This made it possible to determine whether repeated project-month records contain the same forecast amount, different forecast components or a combination of both patterns.

For every group, the query calculated:

- RecordCount, representing the number of source rows;
- TotalForecast, representing the sum of the group;
- AllRows, retaining the underlying records for audit; and
- DistinctAmountCount, representing the number of distinct non-null forecast amounts.

Only groups satisfying the following condition were retained:

```
RecordCount > 1
```

13.2 Power Query Implementation

The following implementation counts the distinct forecast amounts inside the nested **AllRows** table of every repeated business key:

```
let
    Source = Fact_ProjectAccrual,

    GroupedRows =
        Table.Group(
            Source,
            {
                "BusinessUnitSource",
                "AccrualBasis",
                "ProjectID",
                "ForecastYear",
                "ForecastMonthNumber"
            },
            {
                {
                    "RecordCount",
                    each Table.RowCount(_),
                    Int64.Type
                },
                {
                    "TotalForecast",
                    each List.Sum(
```

```

        List.RemoveNulls(
            [ProjectAccrualForecast]
        )
    ),
    type nullable number
},
{
    "AllRows",
    each _,
    type table
}
},
),

FilteredRows =
    Table.SelectRows(
        GroupedRows,
        each [RecordCount] > 1
    ),

AddedDistinctAmountCount =
    Table.AddColumn(
        FilteredRows,
        "DistinctAmountCount",
        each
            List.Count(
                List.Distinct(
                    List.RemoveNulls(
                        Table.Column(
                            [AllRows],
                            "ProjectAccrualForecast"
                        )
                    )
                )
            ),
        Int64.Type
    )

in
    AddedDistinctAmountCount

```

13.3 Repeated Business-Key Results

The completed control produced the following results:

Repeated Business-Key Metric	Result
Repeated project-month business-key groups	10
Source records represented	32
Records above a one-row-per-key grain	22
Groups containing one distinct amount	4
Groups containing multiple distinct amounts	6
Exact duplicate excess rows within the repeated keys	12
Errors in distinct-amount calculation	0

The value of 22 represents the number of records above a theoretical one-row-per-business-key grain. It is not a confirmed duplicate count, because some repeated keys contain different forecast amounts that may represent legitimate forecast components.

The 12 exact duplicate excess rows can also be reconciled within this control by summing:

```
RecordCount - DistinctAmountCount
```

This result agrees with the independent exact-duplicate control described in Section 12.

13.4 Detailed Findings

ProjectID	Forecast Period	Record Count	Distinct Amounts	Interpretation
L9109051	September–December	3 per month	1 per month	The same forecast amount is repeated three times in each month.
L9109067	September–December	4 per month	3 per month	Each group contains multiple forecast components and one repeated amount.
07401158	September	2	2	Two different forecast amounts are recorded for the same project-month.
P8202082	September	2	2	Two different forecast amounts are recorded for the same project-month.

A repeated business key does not automatically represent a duplicate.

A **DistinctAmountCount** of one indicates that the same amount is repeated within the project-month group. A value greater than one indicates that the group contains multiple forecast amounts.

The L9109067 groups require particular attention because each group contains four records and three distinct amounts. These groups contain both multiple forecast components and an exact repeated amount.

No records were removed automatically. All repeated records remain available for Finance validation and reconciliation with the original forecasting logic.

13.5 Repeated-Key Status Classification

An explicit status field was added to make the repeated-key findings easier to review and communicate. The new field was named:

RepeatedKeyStatus

The implemented classification logic was:

```

if [DistinctAmountCount] = 1 then
    "Same Amount Repeated"
else if [RecordCount] > [DistinctAmountCount] then
    "Mixed - Components and Exact Duplicate"
else
    "Multiple Distinct Components"

```

The completed classification produced the following results:

Repeated-Key Status	Groups	Source Rows	Exact Excess Rows
Same Amount Repeated	4	12	8
Mixed – Components and Exact Duplicate	4	16	4
Multiple Distinct Components	2	4	0
Total	10	32	12

The four **Same Amount Repeated** groups relate to project L9109051. Each month contains three identical amounts and therefore two excess exact-repeat rows.

The four **Mixed** groups relate to project L9109067. Each month contains four records but only three distinct amounts. Consequently, each group contains multiple financial components and one exact-repeat row.

The remaining two groups, relating to projects 07401158 and P8202082, contain two records and two distinct amounts. They contain no exact duplicate amount and are more likely to represent separate forecast components, subject to business confirmation.

Repeated-Key Control Result

The exact-duplicate control identified 8 exact duplicate groups involving 20 source records and 12 excess duplicate records.

The broader business-key control identified 10 repeated project-month groups involving 32 source records. Six groups contain multiple forecast amounts and therefore may represent legitimate forecast components rather than simple duplication.

The completed status classification divides these findings into 4 same-amount groups, 4 mixed groups and 2 multiple-distinct-component groups. This confirms that the 12 exact excess rows are concentrated in the first two categories.

Inference – Repeated-Key Classification

The repeated-key analysis proves that repeated project-month records do not all have the same business meaning. Four groups are pure same-amount repeats, four are mixed groups containing both distinct components and an exact repeat, and two contain only distinct components. This evidence supports targeted Finance review instead of deleting all 22 rows above a one-row-per-key grain.

14. CRM Opportunity Preparation and Quality Controls

The third source-preparation phase started with a full-column profile of the open CRM opportunities in `Fact_Opportunities`. The fact query continues to reference the raw staging query:

```
= stg_Opportunities_Raw
```

No business transformations, row deletions or automatic replacements were applied during this profiling step.

14.1 Profiling Configuration

The following Power Query profiling controls were enabled:

- Column quality;
- Column distribution; and
- Column profile.

Profiling was explicitly changed from the default first 1,000 rows to:

```
Column profiling based on entire data set
```

The source contains 126 records and 14 columns.

14.2 Observed CRM Source Structure

Source Column	Observed Type	Required Analytical Role
Status	Text	Opportunity status
Est. Closing Date	Any / date-like	Estimated closing date
Closing Date	Any / date-like	Current closing date
Program start (simulated)	Any / date-like	Simulated program start date
Interest / Opportunity	Text	Opportunity identifier or description
Expected Accrued Net Sales for This Year %	Whole Number	Current-year allocation percentage
Expected Accrued Net Sales for Next Year %	Whole Number	Next-year allocation percentage
% Probability	Percentage	Sales probability
Value in EUR	Decimal Number	Source opportunity value
Est. Value	Decimal Number	Estimated opportunity value
Weighted Est. Value_	Decimal Number	Probability-weighted value
Weighted Est. Value this year	Decimal Number	Current-year weighted value
Business Area	Text	Business-area classification
P&LS BU	Text	Business Unit classification

14.3 Profiling Results

CRM Profiling Metric	Observed Result
Total records	126
Total source columns	14
Profiling scope	Entire dataset
Status	100% valid, 0% error, 0% empty and 1 distinct value
Status value represented	Open
Interest / Opportunity	100% valid, 126 distinct and 126 unique values
Date-like source fields	100% populated in the profile but not yet explicitly typed as Date
Current-year allocation percentage	67% valid and 33% empty
Next-year allocation percentage	68% valid and 32% empty
Probability	98% valid, 2% empty and 11 distinct values
Opportunity and weighted-value fields	100% valid, 0% error and 0% empty
Business Area	100% valid and 1 distinct value
P&LS BU	100% valid and 3 distinct values

The preview includes closing and program-start dates extending into 2027 and 2028. Consequently, the financial-year allocation logic cannot be inferred from the closing date alone and must be validated against the explicit current-year and next-year percentage fields.

The source fields **Value in EUR** and **Est. Value** appear aligned in the preview, but their equivalence must be tested across all 126 records before either field is treated as redundant. Likewise, the weighted-value fields must be reconciled against estimated value and probability rather than accepted without calculation control.

Inference – CRM Source Profiling

The CRM source is structurally compact and contains no visible technical error values, but it is not yet ready for the semantic model. Date-like fields require explicit Date types, Probability contains missing values, and annual allocation percentages are intentionally sparse. The unique opportunity descriptions provide a candidate row identifier, while the three Business Unit values can support cross-source BU standardization. Missing allocation values must be classified through business rules rather than replaced with zero automatically.

14.4 CRM Data-Type Standardization

After the full-dataset profile was reviewed, explicit Power Query data types were assigned to all 14 columns in `Fact_Opportunities`. This step was completed before text cleaning, analytical renaming or row-level quality classification so that subsequent calculations operate on stable types.

CRM Column	Assigned Data Type
Status	Text
Est. Closing Date	Date
Closing Date	Date
Program start (simulated)	Date
Interest / Opportunity	Text
Expected Accrued Net Sales for This Year %	Whole Number
Expected Accrued Net Sales for Next Year %	Whole Number
% Probability	Percentage
Value in EUR	Decimal Number
Est. Value	Decimal Number
Weighted Est. Value_	Decimal Number
Weighted Est. Value this year	Decimal Number
Business Area	Text
P&LS BU	Text

Data-Type Validation Results

Data-Type Control Metric	Result
Source records retained	126
Columns assigned explicit types	14
Text columns	4
Date columns	3
Whole-number allocation columns	2
Percentage columns	1
Decimal-number value columns	4
Type-conversion errors	0
Rows removed or nulls replaced	None

The data types were reviewed after conversion and no errors were observed. The existing missing probability and annual-allocation values were deliberately preserved as nulls. They were not converted to zero because a missing value is not necessarily equivalent to a confirmed zero allocation or probability.

Inference – CRM Data-Type Standardization

All 126 CRM opportunity records can be converted to the required analytical types without technical errors or record loss. The three date fields can now support reliable period comparisons, while the probability and financial-value fields can support controlled calculations. Because nulls were preserved, the next quality rules can distinguish genuinely missing business information from valid zero values rather than masking the difference during type conversion.

14.5 CRM Text Standardization and Analytical Renaming

The four CRM text columns were standardized after data-type validation. The Power Query Trim transformation was applied first, followed by Clean, to remove leading or trailing spaces and non-printing characters:

- Status;
- Interest / Opportunity;
- Business Area; and
- P&LS BU.

The corresponding applied steps were retained as Trimmed Text and Cleaned Text. No rows were filtered and no null values were replaced.

All 14 source fields were then renamed to clear analytical names:

Source Name	Analytical Name
Status	OpportunityStatus
Est. Closing Date	EstimatedClosingDate
Closing Date	ClosingDate
Program start (simulated)	SimulatedProgramStartDate
Interest / Opportunity	OpportunityName
Expected Accrued Net Sales for This Year %	CurrentYearAllocationPct
Expected Accrued Net Sales for Next Year %	NextYearAllocationPct
% Probability	Probability
Value in EUR	OpportunityValue
Est. Value	EstimatedValue
Weighted Est. Value_	WeightedEstimatedValue
Weighted Est. Value this year	CurrentYearWeightedEstimatedValue
Business Area	BusinessArea
P&LS BU	BusinessUnitSource

The renaming was recorded in the Renamed Columns step. The operation changed only metadata: all 126 rows and all 14 source fields remained present.

Inference – CRM Text Standardization and Renaming

Text cleaning reduces false category differences caused by invisible characters or extra spaces. The analytical naming convention also aligns the CRM fact with the existing facts, particularly through the shared `BusinessUnitSource` field. The CRM source is therefore easier to audit, join and explain without changing any source financial value.

14.6 CRM Opportunity Data-Quality Classification

A row-level quality field named `OpportunityQualityStatus` was added after the analytical renaming step. The rule gives priority to missing core identifiers and invalid ranges, followed by financial and completeness review conditions:

```
if [OpportunityStatus] = null
  or Text.Trim(Text.From([OpportunityStatus])) = "" then
  "Critical - Missing Status"

else if [OpportunityName] = null
  or Text.Trim(Text.From([OpportunityName])) = "" then
  "Critical - Missing Opportunity"

else if [BusinessUnitSource] = null
  or Text.Trim(Text.From([BusinessUnitSource])) = "" then
  "Critical - Missing BU"

else if [ClosingDate] = null then
  "Critical - Missing Closing Date"

else if [OpportunityValue] = null then
  "Critical - Missing Opportunity Value"

else if [EstimatedValue] = null then
  "Critical - Missing Estimated Value"

else if [Probability] <> null
  and ([Probability] < 0 or [Probability] > 1) then
  "Critical - Invalid Probability"

else if [CurrentYearAllocationPct] <> null
  and (
    [CurrentYearAllocationPct] < 0
    or [CurrentYearAllocationPct] > 100
  ) then
  "Critical - Invalid Current-Year Allocation"

else if [NextYearAllocationPct] <> null
  and (
    [NextYearAllocationPct] < 0
    or [NextYearAllocationPct] > 100
  ) then
  "Critical - Invalid Next-Year Allocation"

else if [CurrentYearAllocationPct] <> null
  and [NextYearAllocationPct] <> null
  and (
    [CurrentYearAllocationPct]
    + [NextYearAllocationPct]
  ) > 100 then
```

```

    "Critical - Allocation Above 100%"

else if [OpportunityValue] < 0
    or [EstimatedValue] < 0 then
    "Review - Negative Value"

else if [OpportunityValue] = 0
    or [EstimatedValue] = 0 then
    "Review - Zero Value"

else if [Probability] = null then
    "Review - Missing Probability"

else if [CurrentYearAllocationPct] = null
    and [NextYearAllocationPct] = null then
    "Review - Allocation Not Provided"

else
    "Valid"

```

The resulting field was explicitly typed as Text. Profiling was performed on the entire dataset after the classification.

Opportunity Quality Status	Record Count	Share of Records
Valid	95	75.4%
Review – Allocation Not Provided	26	20.6%
Review – Zero Value	2	1.6%
Review – Missing Probability	2	1.6%
Critical – Allocation Above 100%	1	0.8%
Total	126	100.0%

The completed classification contains five distinct statuses, zero errors, zero empty classifications and one status occurring in only one record. No record was deleted automatically. Since the rule returns one status per row, the ordering of conditions deliberately exposes the highest-priority issue first; secondary conditions can still be examined through dedicated control queries.

The single **Critical – Allocation Above 100%** record requires source or Finance validation before it contributes to the forecast. The 26 records without current-year or next-year allocations are review items rather than confirmed errors, because opportunities extending beyond the immediate forecast horizon may legitimately have no allocation yet.

Inference – CRM Opportunity Quality Classification

The control confirms that 95 of 126 opportunities are valid under the current rules and that every record receives an auditable classification. Most exceptions relate to allocation completeness rather than structural source failure. Only one record currently violates an explicit allocation rule, while the remaining review records are retained for business confirmation. The CRM fact is therefore technically controlled, but financial calculation checks are still required before it is used in the final net-sales forecast.

14.7 CRM Weighted-Value Reconciliation

The source probability-weighted opportunity amount was independently recalculated to test whether `WeightedEstimatedValue` agrees with `EstimatedValue` multiplied by `Probability`. Three diagnostic fields were added without overwriting the source values.

The independent weighted amount was calculated as:

```
if [EstimatedValue] = null
  or [Probability] = null then
  null
else
  [EstimatedValue] * [Probability]
```

The resulting field was named `CalculatedWeightedEstimatedValue` and typed as Decimal Number.

The source-to-calculated variance was then created:

```
if [WeightedEstimatedValue] = null
  or [CalculatedWeightedEstimatedValue] = null then
  null
else
  [WeightedEstimatedValue]
  - [CalculatedWeightedEstimatedValue]
```

The resulting field was named `WeightedValueVariance` and typed as Decimal Number. Finally, the tolerance-based reconciliation status was added:

```
if [Probability] = null then
  "Not Testable - Missing Probability"
else if [WeightedEstimatedValue] = null then
  "Not Testable - Missing Weighted Value"
else if Number.Abs([WeightedValueVariance]) <= 0.01 then
  "Match"
else
  "Mismatch"
```

The resulting field was named `WeightedValueCheck` and typed as Text. The EUR 0.01 tolerance prevents harmless binary floating-point residuals from being reported as financial mismatches.

Weighted-Value Control Metric	Result
Source records evaluated	126
Columns after adding the three diagnostics	18
WeightedValueCheck errors	0
WeightedValueCheck empty values	0
Distinct reconciliation statuses	3
Match	123
Not Testable – Missing Probability	2
Mismatch	1
Exact zero variances	118
Non-zero residuals accepted within EUR 0.01 tolerance	5
Material mismatch variance	EUR -468,750.00

The variance definition is source weighted value minus independently calculated weighted value. Therefore, the EUR -468,750.00 result indicates that the source weighted amount is EUR 468,750 lower than the probability-based calculation for the affected opportunity. Two other records cannot be tested because their probability is missing; this is a business-data limitation rather than a Power Query error.

The single mismatch must not be corrected automatically. Its opportunity name, Business Unit, estimated value, probability and source weighted value must first be reviewed together. The difference could reflect a source formula issue, a manual override or a business rule that is not represented by the simple `EstimatedValue * Probability` calculation.

Inference – CRM Weighted-Value Reconciliation

The control demonstrates strong calculation consistency: 123 of 126 records match the independent calculation within one cent. Of these, 118 are exact zero variances and five contain only negligible floating-point residuals. Two rows remain explicitly not testable because Probability is missing, and one material mismatch has been isolated for targeted investigation. This approach prevents a single anomaly from being hidden inside an otherwise reliable aggregate.

14.8 Dedicated CRM Weighted-Value Mismatch Control

The earlier reconciliation identified one material `WeightedValueCheck = "Mismatch"` record. The dedicated data-quality reference query originally planned for this exception has now been created

and is retained in the 03_Data_Quality group:

```
DQ_Opportunity_WeightedValueMismatch
```

The query isolates the exception without changing **Fact_Opportunities**. The source-minus-calculated variance remains EUR -468,750.00 as documented in Section 14.7. The record therefore remains a business-validation item rather than an automatically corrected source value.

Inference – Weighted-Value Exception Isolation

Separating the material mismatch into a dedicated control query keeps the analytical fact source-aligned while making the exception easy to review, demonstrate and audit. The technical control is complete; the business meaning of the mismatch still requires confirmation.

14.9 Opportunity Value versus Estimated Value Reconciliation

The next CRM control compared the source fields **OpportunityValue** and **EstimatedValue**. A diagnostic variance field named **OpportunityValueVariance** was added:

```
if [OpportunityValue] = null
    or [EstimatedValue] = null then
    null
else
    [OpportunityValue] - [EstimatedValue]
```

The resulting field was typed as Decimal Number. A second field, **OpportunityValueCheck**, applies the same EUR 0.01 financial tolerance used in the weighted-value control:

```
if [OpportunityValue] = null then
    "Not Testable - Missing Opportunity Value"
else if [EstimatedValue] = null then
    "Not Testable - Missing Estimated Value"
else if [OpportunityValueVariance] = null then
    "Not Testable - Missing Variance"
else if Number.Abs([OpportunityValueVariance]) <= 0.01 then
    "Match"
else
    "Mismatch"
```

The full-dataset profile shows that both diagnostic fields are 100% valid with zero errors and zero empty values. **OpportunityValueCheck** has one distinct status and the value shown across the dataset is **Match**. Consequently, all 126 CRM records pass this reconciliation within the one-cent tolerance.

Opportunity-Value Control Metric	Result
Source records evaluated	126
OpportunityValueVariance valid values	100%
OpportunityValueVariance errors	0
OpportunityValueVariance empty values	0
OpportunityValueCheck valid values	100%
OpportunityValueCheck errors	0
OpportunityValueCheck empty values	0
Distinct OpportunityValueCheck statuses	1
Observed reconciliation status	Match

Opportunity-Value Reconciliation Result

The CRM source fields **OpportunityValue** and **EstimatedValue** reconcile across all 126 records within the EUR 0.01 tolerance. The earlier profiling observation that the two fields appeared aligned has therefore been converted into an explicit row-level control.

Inference – Opportunity Value versus Estimated Value

This result removes one potential ambiguity from the source structure. The two fields currently carry equivalent analytical values, but both are retained at this stage to preserve source lineage. Any later decision to hide or remove one from the semantic model can now be based on evidence rather than assumption.

14.10 Current-Year Weighted-Value Reconciliation

A further control was added to validate the CRM source field **CurrentYearWeightedEstimatedValue**. The purpose is to test whether the current-year weighted amount is consistent with estimated value, probability and the explicit current-year allocation percentage.

Independent Current-Year Weighted Calculation

The independently calculated amount was added as **CalculatedCurrentYearWeightedValue**:

```
if [EstimatedValue] = null
  or [Probability] = null
  or [CurrentYearAllocationPct] = null then
```

```
    null
  else
    [EstimatedValue]
      * [Probability]
      * ([CurrentYearAllocationPct] / 100)
```

The field was typed as Decimal Number. The full-dataset profile shows 67% valid values, 0% errors and 33% empty values. The empty values are preserved because the calculation is intentionally not forced when a required business input is missing.

Current-Year Variance

The source-to-calculated variance was then created as **CurrentYearWeightedVariance**:

```
if [CurrentYearWeightedEstimatedValue] = null
  or [CalculatedCurrentYearWeightedValue] = null then
  null
else
  [CurrentYearWeightedEstimatedValue]
    - [CalculatedCurrentYearWeightedValue]
```

The variance was typed as Decimal Number. The current profile reports:

- 67% valid values;
- 0% errors; and
- 33% empty values.

The empty variance values are expected wherever the source or calculated current-year amount cannot be evaluated.

Null-Safe Current-Year Reconciliation Status

The final row-level control is named **CurrentYearWeightedCheck**. The implemented logic is:

```
if [Probability] = null then
  "Not Testable - Missing Probability"
else if [CurrentYearAllocationPct] = null then
  "Not Testable - Missing Current-Year Allocation"
else if [CurrentYearWeightedEstimatedValue] = null then
  "Not Testable - Missing Source Value"
else if [CalculatedCurrentYearWeightedValue] = null
  or [CurrentYearWeightedVariance] = null then
  "Not Testable - Missing Calculated Value"
else if Number.Abs([CurrentYearWeightedVariance]) <= 0.01 then
  "Match"
else
  "Mismatch"
```

The field was explicitly typed as Text.

Power Query Step-Order Correction

The first version of **CurrentYearWeightedCheck** was created before **CurrentYearWeightedVariance**. Because Power Query evaluates applied steps sequentially, the check could not reliably reference a field that did not yet exist at that point in the query.

The control was corrected without changing the source data:

1. create and type `CalculatedCurrentYearWeightedValue`;
2. create and type `CurrentYearWeightedVariance`;
3. remove the earlier misplaced `CurrentYearWeightedCheck`;
4. recreate `CurrentYearWeightedCheck` after the variance exists;
5. reorder the diagnostic columns for readability; and
6. explicitly type the recreated check as `Text`.

The applied-step sequence now includes the later `Removed Columns`, `Added Custom9`, `Reordered Columns` and `Changed Type11` steps that implement this correction.

Current-Year Control Results at the Present Stage

The completed profile now shows:

Current-Year Weighted Control Metric	Result
Source records retained	126
Columns in Fact__Opportunities after diagnostics	23
CalculatedCurrentYearWeightedValue valid values	67%
CalculatedCurrentYearWeightedValue errors	0
CalculatedCurrentYearWeightedValue empty values	33%
CurrentYearWeightedVariance valid values	67%
CurrentYearWeightedVariance errors	0
CurrentYearWeightedVariance empty values	33%
CurrentYearWeightedCheck valid values	100%
CurrentYearWeightedCheck errors	0
CurrentYearWeightedCheck empty values	0
Distinct CurrentYearWeightedCheck statuses	3
Exact status counts	85 Match / 39 Missing Current-Year Allocation / 2 Missing Probability

The three statuses have now been quantified in a dedicated summary query. Across all 126 CRM opportunity records, 85 are Match, 39 are Not Testable - Missing Current-Year Allocation, and 2 are Not Testable - Missing Probability. No Mismatch row appears in the status summary. Therefore, all 85 records for which the required current-year inputs are available reconcile within the EUR 0.01 tolerance.

Current-Year Weighted Control Result

The current-year reconciliation logic is now technically stable. The variance contains no errors, and the final **CurrentYearWeightedCheck** is 100% valid with zero errors and zero empty classifications across all 126 records. The completed status summary confirms 85 matching records, 39 records that cannot be tested because the current-year allocation is missing, and 2 records that cannot be tested because Probability is missing.

Inference – Current-Year Weighted Reconciliation

The control distinguishes a true numerical mismatch from a record that cannot be tested because a business input is missing. This is important for forecast governance: null allocation or probability values remain visible as explicit “Not Testable” categories rather than being silently converted to zero. The summary shows that 85 of 126 records (67.5%) are fully testable and all 85 reconcile, while 41 records (32.5%) remain not testable because one of the required business inputs is missing. The next technical focus can therefore move to broader allocation completeness and cross-year allocation validation.

14.11 Current-Year Weighted Status Summary Query

A dedicated reference query was created from **Fact_Opportunities** to summarize the final row-level status without changing the analytical fact. The query is named:

DQ_Opportunity_CurrentYearWeightedStatus

The implemented Power Query logic is:

```
let
    Source = Fact_Opportunities,

    GroupedRows =
        Table.Group(
            Source,
            {"CurrentYearWeightedCheck"},
            {
                {
                    "RecordCount",
                    each Table.RowCount(_),
                    Int64.Type
                }
            }
        ),

    SortedRows =
        Table.Sort(
            GroupedRows,
            {
                {"RecordCount", Order.Descending}
            }
        )

in
    SortedRows
```

The completed query returns exactly three rows:

CurrentYearWeightedCheck	Record Count	Share
Match	85	67.5%
Not Testable - Missing Current-Year Allocation	39	31.0%
Not Testable - Missing Probability	2	1.6%
Total	126	100.0%

Current-Year Status Summary Result

The control reconciles exactly to the 126 source opportunity records. There are 85 fully testable records and every one of them is a **Match**. No **Mismatch** status is present. The remaining 41 records are not calculation failures: 39 are missing the current-year allocation and 2 are missing Probability.

Inference – Current-Year Testability

The distinction between a failed reconciliation and an untestable record is now quantified explicitly. The calculation itself is consistent for every record that has the required inputs. The remaining issue is data completeness, not a detected numerical discrepancy. This provides a clear business message for the interview: the model does not convert missing allocation or probability into zero and therefore does not create false forecast certainty.

14.12 Current-Year and Next-Year Allocation Validation

After the current-year weighted-value controls were completed, a dedicated reference query was created to validate the relationship between **CurrentYearAllocationPct** and **NextYearAllocationPct**. The query was named:

```
DQ_Opportunity_AllocationValidation
```

The purpose of the control is to distinguish valid two-year allocations from missing, incomplete or logically inconsistent allocation patterns without modifying **Fact_Opportunities**.

Allocation Total Calculation

A diagnostic field named **AllocationTotalPct** was added. The total is calculated only when both annual allocation values are present:

```
if [CurrentYearAllocationPct] = null
  or [NextYearAllocationPct] = null then
  null
else
  [CurrentYearAllocationPct]
  + [NextYearAllocationPct]
```

The completed profile shows that **AllocationTotalPct** is 59% populated, 41% empty and contains 0 errors. This is consistent with 74 records having both annual allocation values available and 52 records missing at least one of the two inputs.

Allocation Validation Status

A second diagnostic field named **AllocationValidationStatus** was added using the following null-safe classification:

```
if [CurrentYearAllocationPct] = null
  and [NextYearAllocationPct] = null then
  "Review - Both Allocations Missing"

else if [CurrentYearAllocationPct] = null then
  "Review - Missing Current-Year Allocation"

else if [NextYearAllocationPct] = null then
  "Review - Missing Next-Year Allocation"

else if [CurrentYearAllocationPct] < 0
  or [CurrentYearAllocationPct] > 100
  or [NextYearAllocationPct] < 0
  or [NextYearAllocationPct] > 100 then
  "Critical - Allocation Out of Range"

else if [AllocationTotalPct] > 100 then
  "Critical - Allocation Above 100%"

else if [AllocationTotalPct] = 100 then
  "Valid - Allocation Total 100%"

else
  "Review - Allocation Below 100%"
```

The final diagnostic query retains the following review fields:

- OpportunityName;
- BusinessUnitSource;
- ClosingDate;
- SimulatedProgramStartDate;
- CurrentYearAllocationPct;
- NextYearAllocationPct;
- AllocationTotalPct;
- AllocationValidationStatus; and
- OpportunityQualityStatus.

Allocation Validation Results

The full-dataset profile contains 126 records, 0 errors, 0 empty `AllocationValidationStatus` values and 6 observed statuses. The distribution shown by the completed value profile corresponds to:

Allocation Validation Status	Record Count	Share
Valid – Allocation Total 100%	70	55.6%
Review – Both Allocations Missing	29	23.0%
Review – Missing Current-Year Allocation	12	9.5%
Review – Missing Next-Year Allocation	11	8.7%
Review – Allocation Below 100%	3	2.4%
Critical – Allocation Above 100%	1	0.8%
Total	126	100.0%

The profile is internally consistent with the individual allocation-column completeness results. The 29 records with both allocations missing plus the 12 records missing only the current-year allocation produce 41 missing current-year values. Similarly, the same 29 records plus the 11 records missing only the next-year allocation produce 40 missing next-year values.

Among the 74 records where both allocation percentages are present, 70 sum to exactly 100%, 3 sum to less than 100%, and 1 sums to more than 100%. No **Critical – Allocation Out of Range** status is observed in the completed profile.

Allocation Validation Result

The allocation-validation query classifies every CRM opportunity without changing the source fact. Seventy records have a complete two-year allocation that totals exactly 100%. Fifty-two records are missing one or both annual allocation inputs. Four additional records have both inputs present but do not sum to 100%: three are below 100% and one is above 100%.

Inference – Cross-Year Allocation Validation

The main allocation issue is completeness rather than widespread arithmetic inconsistency. Most records with both annual allocation values present reconcile to exactly 100%. The single allocation-above-100% record is a clear business exception, while the three below-100% records require interpretation because their remaining share may belong to a later financial year. Missing allocation values are therefore preserved as missing rather than being replaced by zero. This keeps the forecast transparent and prevents false certainty.

14.13 Allocation Status Summary Query

The visual distribution from the allocation-validation query was converted into an explicit grouped control table using a new reference query:

```
DQ_Opportunity_AllocationStatusSummary
```

The implemented Power Query logic is:

```
let
    Source = DQ_Opportunity_AllocationValidation,

    GroupedRows =
        Table.Group(
            Source,
            {"AllocationValidationStatus"},
            {
                {
                    "RecordCount",
```

```

        each Table.RowCount(_),
        Int64.Type
    }
}
),
SortedRows =
    Table.Sort(
        GroupedRows,
        {
            {"RecordCount", Order.Descending}
        }
    )
in
    SortedRows

```

The grouped output confirms the following exact distribution:

Allocation Validation Status	Record Count
Valid – Allocation Total 100%	70
Review – Both Allocations Missing	29
Review – Missing Current-Year Allocation	12
Review – Missing Next-Year Allocation	11
Review – Allocation Below 100%	3
Critical – Allocation Above 100%	1
Total	126

Inference – Allocation Status Summary

The grouped control independently reproduces the allocation distribution and reconciles exactly to all 126 CRM opportunities. This converts a visual profile into an auditable control table that can later be reused directly in a data-quality dashboard. No allocation record is lost between the source fact, the row-level validation query and the grouped summary.

14.14 Detailed Allocation Exceptions

A dedicated exception query was created from `DQ_Opportunity_AllocationValidation` to isolate only the four records whose complete two-year allocation does not total 100%:

```
DQ_Opportunity_AllocationExceptions
```

The query retains only:

```

AllocationValidationStatus =
    "Review - Allocation Below 100%"
or
AllocationValidationStatus =
    "Critical - Allocation Above 100%"

```

The resulting four records are:

Opportunity	Business Unit	Closing Date	Program Start	CY %	NY %	Total %
Open custom opportunity 111	University Programs BU	2026-08-31	2026-11-29	30	80	110
Open custom opportunity 86	Qualification Programs BU	2026-09-30	2026-12-29	10	40	50
Open custom opportunity 3	ExEd Programs BU	2028-09-30	2028-12-29	25	25	50
Open custom opportunity 33	ExEd Programs BU	2026-12-31	2027-03-31	0	30	30

The first record is the only critical over-allocation: its current-year and next-year shares sum to 110%. Its existing OpportunityQualityStatus is also **Critical - Allocation Above 100%**.

The remaining three records are below 100%. Their existing quality statuses show that two are otherwise **Valid**, while **Open custom opportunity 3** is already classified as **Review - Zero Value**. The date pattern is not uniform: one below-100% record closes and starts in 2028, one closes in late 2026 and starts in early 2027, and one both closes and starts in 2026.

The three below-100% cases must not be automatically increased to 100%. A remaining share could reflect revenue allocated to a later financial year, incomplete source planning, or another business rule not represented by the two available annual fields. Likewise, the 110% record must be validated with the source owner or Finance before any correction is made.

Inference – Allocation Exception Review

The four non-100% cases are heterogeneous rather than one repeated technical error. The 110% record is a clear over-allocation exception. The three below-100% records require business interpretation because their closing and program-start dates span different horizons. In particular, the 2028-dated record supports the possibility that part of the revenue belongs beyond the current and next financial years, whereas the 2026-dated 50% case cannot be explained by date horizon alone. The correct action is targeted review, not blanket normalization.

14.15 Missing Allocation Horizon Analysis

The 52 opportunities with one or both annual allocation fields missing were grouped by allocation status, closing year and simulated program-start year in a dedicated reference query:

```
DQ_Opportunity_MissingAllocationByYear
```

The query preserves the three missing-allocation categories and adds **ClosingYear** and **ProgramStartYear** before grouping.

The completed result is:

Allocation Status	Closing Year	Program Start Year	Record Count
Review – Both Allocations Missing	2024	2024	1
Review – Both Allocations Missing	2025	2025	1
Review – Both Allocations Missing	2026	2026	15
Review – Both Allocations Missing	2026	2027	6
Review – Both Allocations Missing	2027	2027	4
Review – Both Allocations Missing	2027	2028	1
Review – Both Allocations Missing	2028	2028	1
Review – Missing Current-Year Allocation	2026	2026	5
Review – Missing Current-Year Allocation	2026	2027	4
Review – Missing Current-Year Allocation	2027	2027	3
Review – Missing Next-Year Allocation	2026	2026	8
Review – Missing Next-Year Allocation	2026	2027	3
Total			52

The closing-year distribution is:

- 2024: 1 record;
- 2025: 1 record;
- 2026: 41 records;
- 2027: 8 records; and
- 2028: 1 record.

Therefore, 41 of 52 missing-allocation opportunities, or 78.8%, close in 2026. The program-start-year distribution is 1 record in 2024, 1 in 2025, 28 in 2026, 20 in 2027 and 2 in 2028.

Inference – Missing Allocation Horizon

Missing annual allocation is not mainly a long-horizon phenomenon. Nearly four out of five affected opportunities close in 2026, and 28 of the 52 also start their simulated program in 2026. The issue therefore includes a material near-term data-completeness component and cannot be explained simply by the presence of future-dated opportunities. This makes the 2026 records the highest priority for targeted review before forecast aggregation.

14.16 Detailed 2026 Missing-Allocation Review

A second reference query was created to isolate the near-term opportunities identified in the horizon analysis:

```
DQ_Opportunity_2026MissingAllocationDetail
```

The query retains records where:

- **ClosingDate** is in 2026; and
- one or both annual allocation inputs are missing.

The completed query contains exactly 41 rows. The status composition is:

2026 Missing-Allocation Status	Record Count	Share of 41
Both Allocations Missing	21	51.2%
Missing Current-Year Allocation	9	22.0%
Missing Next-Year Allocation	11	26.8%
Total	41	100.0%

Within these 41 near-term records, 28 have a simulated program-start year of 2026 and 13 have a simulated program-start year of 2027. The detail query keeps the opportunity name, Business Unit, closing date, simulated program-start date, both allocation fields, the allocation total and the allocation-validation status visible for direct review.

Inference – 2026 Missing-Allocation Detail

The 2026 detail confirms that the allocation-completeness issue is operationally relevant to the current forecast horizon. More than half of the 41 near-term records are missing both allocation fields, while the remaining records are split between missing current-year and missing next-year allocation. Because these are not merely distant opportunities, the next useful control is to measure their financial exposure rather than only their record count.

14.17 Financial Exposure of 2026 Missing Allocations

The 41 near-term opportunities identified in the 2026 missing-allocation detail were aggregated by missing-allocation status and Business Unit in a dedicated control query:

```
DQ_Opportunity_2026MissingAllocationExposure
```

The completed result is:

Missing Allocation Status	Business Unit	Records	Estimated Value	Weighted Value
Missing Current-Year Allocation	ExEd Programs BU	9	EUR 1,335,562.50	EUR 1,034,593.75
Both Allocations Missing	Qualification Programs BU	3	EUR 1,312,500.00	EUR 615,625.00
Both Allocations Missing	ExEd Programs BU	9	EUR 553,925.00	EUR 195,020.00
Missing Next-Year Allocation	ExEd Programs BU	6	EUR 348,306.25	EUR 171,812.50
Missing Next-Year Allocation	University Programs BU	5	EUR 288,025.00	EUR 156,991.25
Both Allocations Missing	University Programs BU	9	EUR 370,380.48	EUR 143,940.24
Total		41	EUR 4,208,699.23	EUR 2,317,982.74

By missing-allocation status, the same 41 records reconcile to:

- 21 records with both allocations missing: EUR 2,236,805.48 estimated value and EUR 954,585.24 weighted value;
- 9 records missing only current-year allocation: EUR 1,335,562.50 estimated value and EUR 1,034,593.75 weighted value; and
- 11 records missing only next-year allocation: EUR 636,331.25 estimated value and EUR 328,803.75 weighted value.

Inference – 2026 Missing-Allocation Financial Exposure

The allocation-completeness issue is financially material, not merely a record- count issue. The 41 near-term records represent more than EUR 4.2 million of gross estimated pipeline and approximately EUR 2.32 million of probability- weighted pipeline. The missing current-year allocation category is especially important because only 9 records account for more than EUR 1.03 million of weighted value. This supports prioritizing allocation completion by financial exposure rather than treating all missing records equally.

14.18 2026 Current-Year Allocation Risk by Business Unit

A further control was created to focus only on 2026-closing opportunities where `CurrentYearAllocationPct` is missing:

```
DQ_Opportunity_2026CurrentYearAllocationRisk
```

The completed Business Unit summary is:

Business Unit	Records	Estimated Value	Weighted Value
ExEd Programs BU	18	EUR 1,889,487.50	EUR 1,229,613.75
Qualification Programs BU	3	EUR 1,312,500.00	EUR 615,625.00
University Programs BU	9	EUR 370,380.48	EUR 143,940.24
Total	30	EUR 3,572,367.98	EUR 1,989,178.99

The 30 records with missing current-year allocation account for 85.8% of the EUR 2,317,982.74 weighted exposure identified across all 41 2026 missing- allocation records. ExEd Programs BU alone represents approximately 61.8% of the current-year weighted risk, followed by Qualification Programs BU at approximately 30.9% and University Programs BU at approximately 7.2%.

Inference – Current-Year Allocation Risk

The largest unresolved forecast risk is concentrated rather than evenly distributed. Thirty opportunities cannot be reliably allocated into the 2026 forecast because their current-year allocation is missing, and they represent approximately EUR 1.99 million of weighted pipeline. ExEd Programs BU is the highest-priority review area. The appropriate next step is therefore to classify 2026 opportunities by forecast readiness and keep unresolved weighted pipeline visible as at-risk rather than silently treating it as zero or fully included.

14.19 2026 CRM Forecast Readiness by Business Unit

All CRM opportunities with `ClosingDate` in 2026 were classified according to whether their current-year contribution is ready for use, requires business review or cannot yet be calculated reliably. The dedicated control query is:

```
DQ_Opportunity_2026ForecastReadinessSummary
```

The rule applies the following priority:

1. missing Probability;
2. missing current-year allocation;
3. cross-year allocation above 100%;
4. reconciled current-year weighted value; and
5. any remaining current-year reconciliation review.

The completed Business Unit result is:

Forecast Readiness Status	Business Unit	Rows	Weighted Value	Current-Year Value
Not Ready – Missing Current-Year Allocation	ExEd Programs BU	18	EUR 1,229,613.75	–
Not Ready – Missing Current-Year Allocation	Qualification Programs BU	3	EUR 615,625.00	–
Not Ready – Missing Current-Year Allocation	University Programs BU	7	EUR 143,940.24	–
Not Ready – Missing Probability	University Programs BU	2	EUR 0.00	–
Ready – Reconciled Current-Year Value	ExEd Programs BU	60	EUR 1,779,552.50	EUR 527,877.50
Ready – Reconciled Current-Year Value	University Programs BU	13	EUR 365,763.75	EUR 244,772.50
Ready – Reconciled Current-Year Value	Qualification Programs BU	2	EUR 187,125.00	EUR 35,250.00
Review – Cross-Year Allocation Above 100%	University Programs BU	1	EUR 45,000.00	EUR 13,500.00

The eight Business Unit rows reconcile to 106 opportunities closing in 2026. The 75 Ready records represent EUR 5,109,031.25 of estimated value, EUR 2,332,441.25 of probability-weighted value and EUR 807,900.00 of reconciled current-year value.

The 28 records classified as **Not Ready – Missing Current-Year Allocation** represent EUR 3,522,367.98 of estimated value and EUR 1,989,178.99 of probability-weighted pipeline. Two additional records are classified first as **Not Ready – Missing Probability**; they have EUR 50,000.00 of estimated value. Their weighted source amount is zero, but this zero must not be interpreted as a reliable economic valuation because Probability itself is missing.

Inference – 2026 Forecast Readiness

The readiness control separates usable current-year CRM contribution from pipeline that is financially material but not yet forecastable. Seventy-five of 106 2026-closing opportunities, or 70.8%, are ready and contribute EUR 807,900.00 to the current-year CRM forecast. Twenty-eight records carry EUR 1.989 million of weighted pipeline but cannot yet be allocated to 2026. The two missing-Probability records are also unresolved even though their source weighted value is zero. The single 110% allocation record has a calculated EUR 13,500.00 current-year amount, but that amount remains under review rather than being included automatically.

14.20 Forecast Readiness Status Summary

The Business Unit detail was then collapsed into one row per readiness status in:

```
DQ_Opportunity_2026ForecastReadinessStatusSummary
```

The completed grouped result is:

Forecast Readiness Status	Rows	Weighted Value	Current-Year Value
Ready – Reconciled Current-Year Value	75	EUR 2,332,441.25	EUR 807,900.00
Not Ready – Missing Current-Year Allocation	28	EUR 1,989,178.99	–
Not Ready – Missing Probability	2	EUR 0.00	–
Review – Cross-Year Allocation Above 100%	1	EUR 45,000.00	EUR 13,500.00
Total	106	EUR 4,366,620.24	EUR 821,400.00 calculated

The EUR 821,400.00 calculated current-year total consists of EUR 807,900.00 from Ready opportunities plus EUR 13,500.00 from the one over-allocation review record. Only the Ready portion is considered cleared for forecast use at the present technical stage.

The total weighted value in the readiness table is not a complete measure of economic potential because the two records with missing Probability have a source weighted value of zero. Their EUR 50,000.00 estimated value remains visible as an unresolved exposure. Likewise, the EUR 1,989,178.99 weighted pipeline associated with missing current-year allocation must not be added directly to the EUR 807,900.00 current-year forecast because its 2026 allocation is unknown.

Inference – Forecast Readiness Status Summary

The grouped status control provides the first governed bridge between CRM data quality and forecast use. The current technically cleared CRM contribution is EUR 807,900.00. A further EUR 13,500.00 has been calculated but is held for allocation review. EUR 1,989,178.99 remains probability-weighted pipeline at risk because the current-year split is missing, while EUR 50,000.00 of estimated pipeline remains unresolved because Probability is missing. These values have different meanings and must remain in separate forecast buckets.

14.21 Governed 2026 CRM Forecast Bridge

The four forecast-readiness statuses were converted into explicit forecast treatments in a dedicated reference query:

```
DQ_Opportunity_2026ForecastBridge
```

The completed bridge contains four rows and four financially distinct value columns:

Forecast Treatment	Rows	Primary Governed Value	Amount
Include – Ready Current-Year Forecast	75	Ready Current-Year Forecast	EUR 807,900.00
At Risk – Weighted Pipeline Not Allocated to 2026	28	At-Risk Weighted Pipeline	EUR 1,989,178.99
At Risk – Estimated Pipeline Missing Probability	2	Missing-Probability Estimated Value	EUR 50,000.00
Hold – Current-Year Value Under Review	1	Review Current-Year Value	EUR 13,500.00

The query deliberately stores the four values in separate columns: `ReadyCurrentYearForecastValue`, `ReviewCurrentYearValue`, `AtRiskWeightedPipeline` and `MissingProbabilityEstimatedValue`. Nulls are retained in the non-applicable columns.

Inference – Governed CRM Forecast Bridge

The CRM pipeline now has an explicit forecast-treatment layer. EUR 807,900.00 is technically cleared for inclusion in the current-year CRM forecast. EUR 13,500.00 is calculated but held pending allocation review. EUR 1,989,178.99 remains probability-weighted pipeline that cannot yet be allocated into 2026, while EUR 50,000.00 of estimated opportunity value remains unresolved because Probability is missing. These values are intentionally not added together as one forecast measure because they represent different levels of readiness and different financial meanings.

14.22 Cross-Source Business Unit Vocabulary Audit

Before creating the shared Business Unit dimension, the observed Business Unit labels from all three analytical facts were combined and counted in:

```
DQ_BusinessUnit_SourceValues
```

The completed result is:

Source Dataset	BusinessUnitSource	Record Count
Actual Sales	ExEd	346
Actual Sales	Qualification Programs	120
Actual Sales	University Programs	329
CRM Opportunities	ExEd Programs BU	92
CRM Opportunities	Qualification Programs BU	7
CRM Opportunities	University Programs BU	27
Project Accrual	Missing / null	4
Project Accrual	ExEd Programs BU	162
Project Accrual	Qualification Programs BU	43
Project Accrual	University Programs BU	104

The grouped counts reconcile exactly to the fact-table record counts: 795 Actual Sales rows, 126 CRM Opportunity rows and 313 Project Accrual rows.

The audit shows that CRM Opportunities and Project Accrual already use the same three populated labels:

- ExEd Programs BU;
- Qualification Programs BU; and
- University Programs BU.

Actual Sales represents the same three business areas with shorter labels: **ExEd**, **Qualification Programs** and **University Programs**. The four null accrual records are not assigned to any valid Business Unit automatically.

Inference – Business Unit Vocabulary Audit

The cross-source Business Unit mismatch is semantic rather than structural. CRM and Accrual already share one common vocabulary, while Actual Sales uses shorter names for the same three categories. This can be standardized safely through an explicit mapping layer. The four accrual records with missing Business Unit must remain mapped to an **Unassigned / Missing** BU technical member rather than being guessed into one of the three real Business Units. The next modelling step is therefore to create a canonical **DimBusinessUnit** table before modifying fact-table labels.

14.23 Canonical CRM Business Unit Key Mapping

A numeric **BusinessUnitKey** was added to **Fact_Opportunities** while retaining the source **BusinessUnitSource** field:

```
if [BusinessUnitSource] = null
    or Text.Trim(Text.From([BusinessUnitSource])) = "" then
    0
else if [BusinessUnitSource] = "ExEd Programs BU" then
    1
else if [BusinessUnitSource] = "Qualification Programs BU" then
    2
else if [BusinessUnitSource] = "University Programs BU" then
    3
else
    null
```

The field was typed as Whole Number / **Int64.Type**. The completed profile contains 126 rows, 100% valid values, 0 errors, 0 empty values and 3 distinct keys. The source audit reconciles the distribution to 92 records on key 1, 7 records on key 2 and 27 records on key 3. No CRM Opportunity maps to key 0.

Inference – CRM Business Unit Mapping

The CRM fact is now aligned to the same governed Business Unit structure as Actual Sales and Project Accrual. Because every Opportunity has one of the three known Business Units, no CRM record falls into the technical Unassigned member. The original CRM label remains available for lineage while the numeric key provides a stable relationship field.

15. Current Implementation Status

Implementation Item	Status	Result
Source files imported	Completed	Three Excel sources available
Staging queries created	Completed	Loading disabled
Fact references created	Completed	Three analytical facts available
Actual sales transformation	Completed	795 records prepared
Actual quality classification	Completed	No critical issues
Actual project-month validation	Completed	No duplicate keys
Accrual date preparation	Completed	September–December 2026
Accrual Total identification	Completed	One source Total identified
Accrual Total separation	Completed	Control query created
Accrual detail filtering	Completed	313 detail rows retained
Accrual quality classification	Completed	308 valid and 5 exception records
Accrual column standardization	Completed	Eight final analytical columns
Accrual reconciliation	Completed	EUR 0.00 variance
Accrual duplicate investigation	Completed	8 groups and 12 excess records
Repeated business-key investigation	Completed	10 groups containing 32 source rows
Distinct forecast-amount analysis	Completed	4 same-amount and 6 multiple-amount groups
Repeated-key status classification	Completed	4 same, 4 mixed and 2 component groups
CRM opportunity preparation	Completed technically	Transformation, reconciliation, allocation, forecast-readiness and BU-key controls completed
CRM source profiling	Completed	Entire dataset evaluated
CRM data-type standardization	Completed	14 columns typed with 0 conversion errors
CRM text standardization	Completed	Four text columns received Trim and Clean
CRM analytical renaming	Completed	14 source fields renamed consistently
CRM quality classification	Completed	126 classified rows, 0 errors and 0 empty statuses
CRM weighted-value reconciliation	Completed	123 Match, 2 Not Testable and 1 Mismatch
CRM weighted-value mismatch control	Completed technically	Dedicated DQ query created; one material exception remains for business review
Opportunity value reconciliation	Completed	126 records reconcile; one distinct check status: Match
Current-year weighted calculation	Completed	67% populated, 33% null, 0 errors
Current-year weighted variance	Completed	67% populated, 33% null, 0 errors
Current-year weighted check	Completed	100% valid, 0 errors, 0 empty values and

15.1 Final Semantic Relationship Audit

The semantic model was re-checked in the Power BI **Manage relationships** dialog after the Executive Dashboard and field-visibility cleanup were completed. The final relationship inventory is:

Fact Side	Dimension Side	Status	Cardinality	Filter
Fact_ActualSales[BusinessUnitKey]	DimBusinessUnit[BusinessUnitKey]	Active	*:1	Single
Fact_ActualSales[MonthStartDate]	DimDate[Date]	Active	*:1	Single
Fact_ActualSales[ProjectID]	DimProject[ProjectID]	Active	*:1	Single
Fact_Opportunities[BusinessUnitKey]	DimBusinessUnit[BusinessUnitKey]	Active	*:1	Single
Fact_Opportunities[ClosingDate]	DimDate[Date]	Active	*:1	Single
Fact_Opportunities[SimulatedCRMStartDate]	DimDate[Date]	Inactive	*:1	Single
Fact_ProjectAccrual[BusinessUnitKey]	DimBusinessUnit[BusinessUnitKey]	Active	*:1	Single
Fact_ProjectAccrual[ForecastDate]	DimDate[Date]	Active	*:1	Single
Fact_ProjectAccrual[ProjectID]	DimProject[ProjectID]	Active	*:1	Single

Inference – Final Relationship Audit

The final semantic model retains a controlled star-schema pattern. Eight relationships are active and provide the intended Business Unit, Date, CRM closing-date and Project filter paths. The second CRM date role remains deliberately inactive so the same opportunity fact can support an alternative program-start analysis through explicit DAX without introducing ambiguous simultaneous date filtering.

15.2 Executive Dashboard End-to-End Business Unit Validation

After the Forecast Governance table and the remaining executive visuals were added, the governed Business Unit slicer was re-tested against the complete page. The four available selections produced the following control values:

Business Unit	Actual	Accrual	Base Forecast	Ready CRM
ExEd Programs BU	EUR 6.04M	EUR 3.01M	EUR 9.05M	EUR 527.88K
Qualification Programs BU	EUR 4.87M	EUR 1.77M	EUR 6.64M	EUR 35.25K
University Programs BU	EUR 3.72M	EUR 1.79M	EUR 5.52M	EUR 244.77K
Unassigned / Missing BU	–	EUR 42.55K	EUR 42.55K	–

The CRM governance values also remain consistent with the source controls: ExEd carries approximately EUR 1.23M of at-risk weighted pipeline, Qualification approximately EUR 615.63K, University approximately EUR 143.94K, while the EUR 13.50K review amount and EUR 50.00K missing-Probability exposure appear under University Programs BU. The Unassigned member contains only the known EUR 42,550.00 Accrual amount and no fabricated Actual or CRM contribution.

Inference – End-to-End Business Unit Interaction

The complete Executive Dashboard now demonstrates that one conformed Business Unit dimension controls all business-facing financial and CRM-governance visuals consistently. The explicit Unassigned member behaves as a governance exception rather than leaking into a real Business Unit, while the three populated Business Units preserve their expected financial and CRM risk profiles.

15.3 Technical Field Visibility Cleanup

The model was then cleaned for business-facing use. Technical and control fields needed for joins, sorting, reconciliation and data-quality diagnostics remain physically present in the semantic model, but selected fields are hidden from report authors and consumers.

Confirmed examples include technical Business Unit sort/key helpers, hidden calendar sorting fields such as **YearMonth**, the Project surrogate **ProjectKey**, and selected row-quality or diagnostic fields in the fact tables. Business-facing dimensions, descriptive attributes and governed measures remain visible.

Inference – Semantic Field Governance

Hiding technical fields improves usability without weakening auditability. The report author is guided toward governed dimensions and measures, while the model still retains the keys, sort columns and diagnostic evidence required for maintenance, troubleshooting and future extension.

15.4 Current Completion Point and Next Control

The analytical modelling, relationship construction, main DAX measures, Business Unit and Project filter tests, executive KPI layer, monthly forecast view, Business Unit composition, Top-10 Project composition and Forecast Governance table are now complete.

The next technical control is a final financial reconciliation sign-off. Two zero-variance measures should be added:

```
Base Forecast Reconciliation Variance =
    [Base Full-Year Forecast]
    - (
        [Actual Net Sales]
        + [Project Accrual Forecast]
    )
```

```
Ready CRM Inclusion Variance =
    [Forecast Including Ready CRM]
    - (
        [Base Full-Year Forecast]
        + [CRM Ready Current-Year Value]
    )
```

Both controls are expected to return EUR 0.00 at the all-Business-Unit level and under valid Business Unit selections. After this sign-off, the remaining work is presentation polish, resetting slicers to the intended default state, saving the final checkpoint and preparing the short interview narrative.

Inference – Final Technical Readiness

The solution has moved from data preparation into final assurance. No new forecasting logic is required for the core case; the remaining technical task is to prove that the already validated components reconcile algebraically in the finished report context. A zero-variance sign-off provides a compact final control before presentation.

16. Governance and Process Value

The implemented approach provides:

- transparent source-to-report lineage;
- repeatable transformations during every refresh;
- explicit row-level data-quality controls;
- documented handling of exceptional financial values;
- validation of the intended analytical grain;
- separation of raw, analytical and control queries;
- controlled handling of source Total rows;
- prevention of accidental forecast double-counting;
- distinction between exact duplicates and legitimate forecast components;
- explicit CRM data types with zero conversion errors;
- CRM text cleaning, standardized analytical names and complete row-level quality classification;
- independent weighted-value recalculation with an explicit financial tolerance;
- explicit reconciliation of `OpportunityValue` and `EstimatedValue`;
- independent current-year weighted-value calculation using probability and annual allocation;
- null-safe distinction between numerical mismatches and records that cannot yet be tested because a business input is missing;
- documented correction of Power Query dependency order without changing source data;
- a dedicated current-year weighted-status summary that reconciles all 126 opportunities and separates 85 testable matches from 41 records with missing required inputs;
- explicit cross-year allocation validation that separates complete 100% allocations, incomplete allocation data and true over-allocation exceptions without imputing missing values;
- an auditable grouped allocation-status summary that reconciles all 126 opportunities;
- targeted isolation of four non-100% allocation records together with closing and simulated program-start dates for business interpretation;
- horizon analysis of all 52 missing-allocation records, showing that 41 close in 2026;
- a dedicated 41-row near-term review set for missing-allocation opportunities;
- financial-exposure quantification for the 41 near-term allocation gaps;

- a Business Unit risk summary for the 30 opportunities that cannot yet be allocated reliably into the current financial year;
- a 2026 forecast-readiness classification across all 106 near-term CRM opportunities;
- an auditable four-status summary separating EUR 807,900.00 of Ready current-year CRM value from review and unresolved pipeline exposures;
- a governed CRM forecast bridge that keeps forecast inclusion, held value and unresolved pipeline in separate financial columns;
- a cross-source Business Unit vocabulary audit covering all three analytical facts;
- a governed `DimBusinessUnit` with stable numeric keys and an explicit unassigned member;
- preservation of source Business Unit labels alongside canonical keys for lineage;
- completed canonical BU-key mapping for all 795 Actual Sales records;
- completed canonical BU-key mapping for all 313 Project Accrual records, including 4 explicit Unassigned rows;
- completed canonical BU-key mapping for all 126 CRM Opportunity records;
- removal of staging and DQ helper tables from the visible semantic model without deleting their Power Query controls;
- three active one-to-many Business Unit relationships with single-direction filtering and no fact-to-fact relationship;
- explicit DAX-based validation that the shared Business Unit dimension propagates filters correctly to all three facts;
- exact reconciliation of Business Unit row counts to 795 Actual, 313 Accrual and 126 CRM Opportunity records;
- a complete, unique and error-free 2026 Date dimension with daily, monthly and quarterly attributes;
- two active Single-direction Date relationships from the shared calendar to Actual Sales and Project Accrual;
- explicit monthly validation that January–August contains only Actual rows while September–December contains only Accrual rows at the current source coverage;
- chronological MonthName sorting governed by the numeric MonthNumber;
- explicit DAX measures for Actual Net Sales, Project Accrual Forecast and the combined Base Full-Year Forecast;
- exact source-level reconciliation of the base financial totals to EUR 14,635,067.81 Actual + EUR 6,616,025.78 Accrual = EUR 21,251,093.59;
- formal designation of `DimDate` as the model Date table;
- financial-value validation by Business Unit, including the explicit EUR 42,550.00 Unassigned accrual amount;
- a governed Ready-CRM DAX layer that includes only 75 technically cleared opportunities and

EUR 807,900.00 of reconciled current-year value;

- a current governed annual forecast of EUR 22,058,993.59 after Ready CRM is added to the source-aligned base forecast;
- semantic CRM governance measures that keep Review, At-Risk and Missing-Probability exposures separate from the cleared forecast;
- an optional review-approved scenario of EUR 22,072,493.59 without automatically including unresolved pipeline;
- a disconnected forecast-bucket presentation dimension that introduces no additional fact-filter path;
- a complete 2024–2028 CRM calendar and explicit active/inactive role-playing relationships for Closing Date and Program Start;
- DAX validation proving that both CRM date roles reconcile to all 126 opportunities while retaining their different business timing;
- a governed Project dimension containing 252 clean, unique ProjectIDs;
- two active Single-direction Project relationships from the dimension to Actual Sales and Project Accrual using the validated natural ProjectID;
- Project-level reconciliation proving the new dimension preserves all 795 Actual rows, 313 Accrual rows and the complete EUR 21,251,093.59 base forecast;
- deliberate exclusion of CRM from the Project relationship path because the supplied opportunity source does not contain a confirmed ProjectID mapping;
- an interactive Project slicer test proving that one ProjectID filters both project-based facts without changing the independent CRM governance layer;
- a Top-10 Project forecast ranking that converts the validated Project dimension into an executive prioritization view based on the full-year base forecast; and
- a foundation for reconciliation, Business Unit harmonization, shared time filtering, controlled CRM integration, Project analysis and exception reporting.

17. Next Implementation Phases

The implementation sequence identified earlier is preserved below. Completed items are marked as completed rather than removed from the report:

1. validate the 12 excess exact-duplicate records with Finance (*business validation pending*);
2. validate the four records with missing Business Unit values (*business validation pending*);
3. investigate the single CRM weighted-value mismatch and the two records with missing Probability (*technical isolation completed; business validation pending*);
4. reconcile OpportunityValue with EstimatedValue (*completed: all 126 records are Match within EUR 0.01*);
5. validate current-year and next-year allocation percentages (*technical control completed: 70 records total 100%, 29 have both allocations missing, 12 miss current-year allocation, 11 miss next-year allocation, 3 are below 100%, and 1 is above 100%; business interpretation remains*

open);

6. summarize the exact record counts of the three **CurrentYearWeightedCheck** statuses (*completed: 85 Match, 39 Missing Current-Year Allocation, 2 Missing Probability*);
7. summarize the six allocation-validation statuses (*completed: 70, 29, 12, 11, 3 and 1 records*);
8. isolate and review the four complete-allocation exceptions (*completed technically: totals of 110%, 50%, 50% and 30%; business interpretation pending*);
9. analyze the 52 records with one or both annual allocations missing by closing year and simulated program-start year (*completed: 41 close in 2026, 8 in 2027, 1 in 2028 and 2 in 2024–2025*);
10. isolate the near-term 2026 missing-allocation records (*completed: 41 rows, comprising 21 both missing, 9 missing current-year and 11 missing next-year allocation*);
11. quantify the financial exposure of the 41 2026 missing-allocation records using estimated value and probability-weighted value (*completed: EUR 4,208,699.23 estimated value and EUR 2,317,982.74 weighted value*);
12. isolate 2026 current-year allocation risk by Business Unit (*completed: 30 records, EUR 3,572,367.98 estimated value and EUR 1,989,178.99 weighted value*);
13. classify 2026 CRM opportunities by forecast readiness and quantify reconciled current-year value versus unresolved pipeline exposure (*completed: 75 Ready, 28 Missing Current-Year Allocation, 2 Missing Probability and 1 Cross-Year Allocation Review*);
14. summarize the 2026 forecast-readiness statuses (*completed: EUR 807,900.00 Ready current-year value, EUR 13,500.00 review value and EUR 1,989,178.99 unresolved weighted pipeline*);
15. create a governed 2026 CRM forecast bridge that keeps Ready forecast, Review value and unresolved pipeline exposure separate (*completed: EUR 807,900.00 Ready, EUR 13,500.00 Hold, EUR 1,989,178.99 weighted risk and EUR 50,000.00 missing-Probability exposure*);
16. audit Business Unit source labels across all three analytical facts (*completed: Actual Sales uses 3 shorter labels; CRM and Accrual share 3 canonical labels; 4 Accrual rows remain missing*);
17. create the canonical **DimBusinessUnit** table with an explicit **Unassigned** / **Missing** BU member (*completed: keys 1–3 for the canonical BUs and key 0 for Unassigned*);
18. add the canonical **BusinessUnitKey** to **Fact_ActualSales** while retaining **BusinessUnitSource** (*completed: all 795 rows mapped with 0 errors and 0 empty keys*);
19. add the canonical **BusinessUnitKey** to **Fact_ProjectAccrual**, including key 0 for the 4 missing-BU rows (*completed: 313 rows, 100% valid and 4 distinct keys*);
20. add the canonical **BusinessUnitKey** to **Fact_Opportunities** (*completed: 126 rows, 100% valid and 3 distinct keys*);
21. disable model load for staging and DQ queries while retaining their transformation and audit logic (*completed: only the dimension and three facts remain visible*);
22. standardize Business Unit filtering across all datasets using the canonical dimension (*relationship structure completed: three active 1:* Single-direction relationships*);
23. validate Business Unit relationship filter propagation with row-count measures (*completed:*

ExEd 346/162/92, Qualification 120/43/7, University 329/104/27 and Unassigned 0/4/0 across Actual/Accrual/CRM);

24. create the shared 2026 DimDate (*completed technically: 365 unique dates, 12 months and 4 quarters*);
25. move DimDate from 01_Staging to 04_Dimensions (*completed*);
26. create Date relationships from DimDate[Date] to Fact_ActualSales[MonthStart] and Fact_ProjectAccrual[ForecastMonth] (*completed: both active, many-to-one from fact perspective, Single cross-filter direction*);
27. validate monthly Date-filter propagation across Actual Sales and Project Accrual (*completed: 795 Actual rows in January–August and 313 Accrual rows in September–December*);
28. sort DimDate[MonthName] by MonthNumber (*completed: January–December chronological order confirmed*);
29. mark DimDate as the model Date table using DimDate[Date] (*completed*);
30. create the first financial DAX measures for Actual Net Sales, Project Accrual Forecast and their combined base full-year forecast (*completed: EUR 14,635,067.81 + EUR 6,616,025.78 = EUR 21,251,093.59*);
31. validate the base financial measures by DimBusinessUnit[BusinessUnitName] (*completed: BU rows reconcile exactly to the annual totals*);
32. create governed Ready-CRM measures and validate them by Business Unit (*completed: 75 opportunities and EUR 807,900.00 current-year value*);
33. create Forecast Including Ready CRM (*completed: EUR 22,058,993.59*);
34. surface the remaining CRM governance buckets as semantic measures: EUR 13,500.00 Review current-year value, EUR 1,989,178.99 At-Risk weighted pipeline and EUR 50,000.00 Missing-Probability estimated exposure (*immediate next technical control*);

Status update: completed. The three measures are validated and the optional review-approved scenario equals EUR 22,072,493.59.

35. define and validate the CRM monthly date-role strategy before placing current-year CRM value on the shared Date axis;

Status update: completed. DimCRMDate covers 2024–2028; Closing Date is active, Program Start is inactive, and both roles reconcile to 126 records through DAX validation.

36. create the Project dimension;

Status update: completed. DimProject contains 252 clean, distinct and unique ProjectIDs and is governed under 04_Dimensions.

37. implement and validate Project relationships;

*Status update: completed. Actual Sales and Project Accrual use active Single-direction *:1 relationships to DimProject[ProjectID]; totals reconcile to 795 Actual rows, 313 Accrual rows and EUR 21,251,093.59.*

38. continue the Power BI semantic model and relationships;

Status update: the core Business Unit, Date, CRM Date-role and Project relationship layers are now technically complete.

39. create forecast scenarios and governed CRM-inclusion measures;

Status update: completed for the current case, including Ready CRM, Review, At-Risk and Missing-Probability treatments.

40. complete the final interactive Project filter validation;

Status update: completed. Selecting 07901009 produces EUR 439,132.89 Actual Net Sales + EUR 308,913.00 Project Accrual Forecast = EUR 748,045.89 Base Full-Year Forecast, while CRM remains outside the Project path.

41. create the Top-10 Project forecast-ranking visual;

Status update: completed. The report now contains a descending horizontal ranking of the ten projects with the largest Base Full-Year Forecast.

42. build executive, detailed and data-quality dashboard pages (current immediate implementation phase: Top-10 Actual-versus-Accrual composition and final presentation polish);

Status update: the Top-10 composition visual, Executive KPI layer, monthly-chart correction and four-member Business Unit slicer validation are completed. The immediate next dashboard visual is Business Unit forecast composition.

43. document the future automated architecture; and

44. prepare the final 15-minute interview presentation.

17.1 Immediate Weighted-Value Mismatch Investigation

The step originally documented as the immediate mismatch investigation has now been completed technically. The planned control query has been created as a reference from Fact_Opportunities:

```
DQ_Opportunity_WeightedValueMismatch
```

The control retains only records where:

```
WeightedValueCheck = "Mismatch"
```

The reconciliation documented in Section 14.7 identifies one record with a source-minus-calculated variance of EUR -468,750.00. The following fields remain the review set for business validation:

- OpportunityName;
- BusinessUnitSource;
- ClosingDate;
- OpportunityQualityStatus;
- OpportunityValue;
- EstimatedValue;
- Probability;
- WeightedEstimatedValue;

- CalculatedWeightedEstimatedValue;
- WeightedValueVariance; and
- WeightedValueCheck.

The query is diagnostic only. The source value remains unchanged until the underlying opportunity and its business calculation are confirmed.

17.2 Outstanding Business Validation

The following exceptions remain open for business confirmation even though the corresponding technical controls have been implemented:

Business Validation Item	Current Result
Exact duplicate excess rows	12 rows requiring Finance confirmation
Potential duplicate overstatement	EUR 175,318.63 diagnostic estimate
Missing Business Unit classification	4 rows with EUR 42,550.00 forecast value
CRM weighted-value mismatch	1 row with EUR -468,750.00 source-minus-calculated variance
CRM records not testable for weighted value	2 rows with missing Probability
CRM allocation above 100%	Open custom opportunity 111: 30% + 80% = 110%
CRM allocation below 100%	3 records: totals of 50%, 50% and 30% requiring business interpretation
Below-100% date patterns	2026/2026, 2028/2028 and 2026/2027 closing/program-start combinations
CRM records missing both annual allocations	29 records
CRM records missing only current-year allocation	12 records
CRM records missing only next-year allocation	11 records
Missing-allocation records closing in 2026	41 of 52 records (78.8%)
2026 missing-allocation composition	21 both missing, 9 missing current-year and 11 missing next-year
2026 missing-allocation weighted exposure	EUR 2,317,982.74 across 41 records
2026 missing-current-year-allocation weighted risk	EUR 1,989,178.99 across 30 records
Largest current-year allocation risk BU	ExEd Programs BU: EUR 1,229,613.75 weighted value across 18 records
2026 Ready CRM opportunities	75 records with EUR 807,900.00 reconciled current-year value
2026 unresolved current-year allocation pipeline	28 records with EUR 1,989,178.99 weighted value
2026 missing-Probability exposure	2 records with EUR 50,000.00 estimated value; source weighted value is not reliable
2026 cross-year allocation review value	1 record with EUR 13,500.00 calculated current-year value held for review
	EUR 807,900.00 cleared for inclusion

17.3 Completed Current-Year Status Summary

The previously documented immediate technical step has now been completed. A reference query was created from `Fact_Opportunities` and named:

```
DQ_Opportunity_CurrentYearWeightedStatus
```

In Advanced Editor, use:

```
let
    Source = Fact_Opportunities,

    GroupedRows =
        Table.Group(
            Source,
            {"CurrentYearWeightedCheck"},
            {
                {
                    "RecordCount",
                    each Table.RowCount(_),
                    Int64.Type
                }
            }
        ),

    SortedRows =
        Table.Sort(
            GroupedRows,
            {
                {"RecordCount", Order.Descending}
            }
        )

in
    SortedRows
```

The output contains three status rows and the `RecordCount` values sum exactly to 126:

- Match: 85 records;
- Not Testable - Missing Current-Year Allocation: 39 records; and
- Not Testable - Missing Probability: 2 records.

No Mismatch status is present. The query is diagnostic only: no source record was removed or changed. The result means that all 85 records with the required inputs pass the current-year weighted-value reconciliation, while the remaining 41 records are explicitly excluded from numerical testing because an input is missing.

17.4 Completed Allocation Validation

The broader allocation review has now been implemented in `DQ_Opportunity_AllocationValidation`. The control calculates a two-year allocation total only when both inputs are present and classifies all 126 opportunities into six observed statuses. The result separates complete 100% allocations from missing-input cases and from the four records whose two-year total differs from 100%.

No allocation value was imputed, replaced or overwritten. The control remains diagnostic and preserves the source-aligned fact.

17.5 Completed Allocation Status Summary

The visual allocation distribution has now been converted into an explicit grouped control table. A reference query from `DQ_Opportunity_AllocationValidation` was created and named:

```
DQ_Opportunity_AllocationStatusSummary
```

In Advanced Editor, use:

```
let
    Source = DQ_Opportunity_AllocationValidation,

    GroupedRows =
        Table.Group(
            Source,
            {"AllocationValidationStatus"},
            {
                {
                    "RecordCount",
                    each Table.RowCount(_),
                    Int64.Type
                }
            }
        ),

    SortedRows =
        Table.Sort(
            GroupedRows,
            {
                {"RecordCount", Order.Descending}
            }
        )

in
    SortedRows
```

The completed output contains six status rows whose `RecordCount` values sum exactly to 126. The confirmed counts are 70, 29, 12, 11, 3 and 1 in descending order. This grouped query provides an auditable count table that can be used directly in the report and later in a data-quality dashboard.

Inference – Grouped Allocation Control

The grouped status table reconciles exactly to the source opportunity count and therefore confirms that the row-level allocation classification is complete. It also provides a compact control table suitable for future dashboard KPI cards and exception monitoring.

17.6 Completed Allocation Exception Isolation

The four complete-allocation records whose two-year total differs from 100% have now been isolated in:

```
DQ_Opportunity_AllocationExceptions
```

The completed exception set contains one 110% over-allocation and three below-100% totals of 50%, 50% and 30%. Closing and simulated program-start dates have been retained for each exception to support business interpretation.

Inference – Exception Isolation

The exceptions are few and individually reviewable. Their date patterns differ, so a single automatic correction rule would be unsafe. The dedicated exception query keeps the fact table unchanged while making the four records easy to demonstrate and validate with Finance.

17.7 Completed Missing Allocation Horizon Analysis

The 52 records where one or both annual allocation inputs are missing have now been analyzed by closing year and simulated program-start year. The purpose was to determine whether missing allocation was concentrated in distant opportunities rather than to fill the missing values.

Create a new reference query from `DQ_Opportunity_AllocationValidation` and name it:

```
DQ_Opportunity_MissingAllocationByYear
```

In Advanced Editor, use:

```
let
    Source = DQ_Opportunity_AllocationValidation,

    FilteredRows =
        Table.SelectRows(
            Source,
            each
                [AllocationValidationStatus] =
                    "Review - Both Allocations Missing"
                or [AllocationValidationStatus] =
                    "Review - Missing Current-Year Allocation"
                or [AllocationValidationStatus] =
                    "Review - Missing Next-Year Allocation"
        ),

    AddedClosingYear =
        Table.AddColumn(
            FilteredRows,
            "ClosingYear",
            each
                if [ClosingDate] = null
                then null
                else Date.Year([ClosingDate]),
            Int64.Type
        ),

    AddedProgramStartYear =
        Table.AddColumn(
            AddedClosingYear,
            "ProgramStartYear",
            each
                if [SimulatedProgramStartDate] = null
                then null
                else Date.Year([SimulatedProgramStartDate]),
            Int64.Type
        ),

    GroupedRows =
        Table.Group(
            AddedProgramStartYear,
            {
```

```

        "AllocationValidationStatus",
        "ClosingYear",
        "ProgramStartYear"
    },
    {
        {
            "RecordCount",
            each Table.RowCount(_),
            Int64.Type
        }
    }
),

SortedRows =
    Table.Sort(
        GroupedRows,
        {
            {"AllocationValidationStatus", Order.Ascending},
            {"ClosingYear", Order.Ascending},
            {"ProgramStartYear", Order.Ascending}
        }
    )

in
    SortedRows

```

The completed grouped result reconciles exactly to 52 records. Forty-one of them close in 2026, which represents 78.8% of the missing-allocation population. Eight close in 2027, one in 2028 and two in 2024–2025.

Inference – Missing Allocation Horizon Control

The result disproves the assumption that missing allocation is mainly caused by long-dated pipeline opportunities. Most affected records are near-term 2026 opportunities and therefore deserve direct forecast-governance attention.

17.8 Completed 2026 Missing-Allocation Detail

A dedicated reference query was then created to isolate the 41 records with **ClosingDate** in 2026 and at least one missing annual allocation:

```
DQ_Opportunity_2026MissingAllocationDetail
```

The completed detail set contains:

- 21 records with both annual allocations missing;
- 9 records missing only the current-year allocation; and
- 11 records missing only the next-year allocation.

The 41-row total reconciles exactly to the 2026 segment identified by the horizon analysis.

Inference – Near-Term Allocation Detail

The near-term review set is now explicit and auditable. Because 41 records are affected, record count alone is no longer sufficient for prioritization; the next control should quantify the financial value represented by these opportunities.

17.9 Completed 2026 Missing-Allocation Financial Exposure

The financial exposure of the 41 near-term missing-allocation records has now been quantified using a reference query from Fact_Opportunities named:

DQ_Opportunity_2026MissingAllocationExposure

In Advanced Editor, use:

```
let
    Source = Fact_Opportunities,

    FilteredRows =
        Table.SelectRows(
            Source,
            each
                [ClosingDate] <> null
                and Date.Year([ClosingDate]) = 2026
                and (
                    [CurrentYearAllocationPct] = null
                    or [NextYearAllocationPct] = null
                )
        ),

    AddedMissingAllocationStatus =
        Table.AddColumn(
            FilteredRows,
            "MissingAllocationStatus",
            each
                if [CurrentYearAllocationPct] = null
                    and [NextYearAllocationPct] = null then
                    "Both Allocations Missing"
                else if [CurrentYearAllocationPct] = null then
                    "Missing Current-Year Allocation"
                else
                    "Missing Next-Year Allocation",
            type text
        ),

    GroupedRows =
        Table.Group(
            AddedMissingAllocationStatus,
            {
                "MissingAllocationStatus",
                "BusinessUnitSource"
            },
            {
                {
                    "RecordCount",
                    each Table.RowCount(_),
                    Int64.Type
                },
                {
                    "TotalEstimatedValue",
                    each List.Sum(
                        List.RemoveNulls([EstimatedValue])
                    ),
                    type nullable number
                }
            }
        )
```

```

        {
            "TotalWeightedEstimatedValue",
            each List.Sum(
                List.RemoveNulls([WeightedEstimatedValue])
            ),
            type nullable number
        }
    }
),

SortedRows =
    Table.Sort(
        GroupedRows,
        {
            {"TotalWeightedEstimatedValue", Order.Descending},
            {"RecordCount", Order.Descending}
        }
    )

in
    SortedRows

```

The completed result reconciles exactly to 41 records. Their combined **EstimatedValue** is EUR 4,208,699.23 and their combined **WeightedEstimatedValue** is EUR 2,317,982.74.

Inference – Financial Exposure Control

The missing-allocation population carries enough financial value to affect the forecast materially. Prioritization should therefore be driven by weighted financial exposure in addition to record counts.

17.10 Completed 2026 Current-Year Allocation Risk

A second control was created from **Fact_Opportunities** to retain only 2026-closing opportunities whose current-year allocation is missing:

```
DQ_Opportunity_2026CurrentYearAllocationRisk
```

The implemented logic is:

```

let
    Source = Fact_Opportunities,

    FilteredRows =
        Table.SelectRows(
            Source,
            each
                [ClosingDate] <> null
                and Date.Year([ClosingDate]) = 2026
                and [CurrentYearAllocationPct] = null
        ),

    GroupedRows =
        Table.Group(
            FilteredRows,
            {
                "BusinessUnitSource"
            },

```

```

    {
        {
            "RecordCount",
            each Table.RowCount(_),
            Int64.Type
        },
        {
            "TotalEstimatedValue",
            each List.Sum(
                List.RemoveNulls([EstimatedValue])
            ),
            type nullable number
        },
        {
            "TotalWeightedEstimatedValue",
            each List.Sum(
                List.RemoveNulls([WeightedEstimatedValue])
            ),
            type nullable number
        }
    }
),

SortedRows =
    Table.Sort(
        GroupedRows,
        {
            {
                "TotalWeightedEstimatedValue",
                Order.Descending
            }
        }
    )

in
    SortedRows

```

The completed result contains 30 records in total and EUR 1,989,178.99 of probability-weighted pipeline whose current-year allocation is not yet available. ExEd Programs BU contributes EUR 1,229,613.75 of this amount, Qualification Programs BU EUR 615,625.00 and University Programs BU EUR 143,940.24.

Inference – 2026 Current-Year Allocation Risk Control

The unresolved current-year allocation issue is concentrated in ExEd Programs BU and represents a material portion of the near-term CRM pipeline. These opportunities should remain visible as an explicit at-risk bucket until Finance or the opportunity owner confirms the annual split.

17.11 Completed 2026 Opportunity Forecast Readiness

All CRM opportunities closing in 2026 have now been classified according to whether their current-year forecast contribution can be used, requires review, or cannot yet be calculated reliably.

Create a new reference query from **Fact_Opportunities** and name it:

```
DQ_Opportunity_2026ForecastReadinessSummary
```

In Advanced Editor, use:

```
let
    Source = Fact_Opportunities,

    FilteredRows =
        Table.SelectRows(
            Source,
            each
                [ClosingDate] <> null
                and Date.Year([ClosingDate]) = 2026
        ),

    AddedForecastReadinessStatus =
        Table.AddColumn(
            FilteredRows,
            "ForecastReadinessStatus",
            each
                if [Probability] = null then
                    "Not Ready - Missing Probability"
                else if [CurrentYearAllocationPct] = null then
                    "Not Ready - Missing Current-Year Allocation"
                else if [OpportunityQualityStatus] =
                    "Critical - Allocation Above 100%" then
                    "Review - Cross-Year Allocation Above 100%"
                else if [CurrentYearWeightedCheck] = "Match" then
                    "Ready - Reconciled Current-Year Value"
                else
                    "Review - Current-Year Reconciliation",
            type text
        ),

    GroupedRows =
        Table.Group(
            AddedForecastReadinessStatus,
            {
                "ForecastReadinessStatus",
                "BusinessUnitSource"
            },
            {
                {
                    "RecordCount",
                    each Table.RowCount(_),
                    Int64.Type
                },
                {
                    "TotalEstimatedValue",
                    each List.Sum(
                        List.RemoveNulls([EstimatedValue])
                    ),
                    type nullable number
                },
                {
                    "TotalWeightedEstimatedValue",
                    each List.Sum(
                        List.RemoveNulls([WeightedEstimatedValue])
                    ),
                    type nullable number
                }
            },
        )
```

```

        {
            "TotalCalculatedCurrentYearValue",
            each
                let
                    Values =
                        List.RemoveNulls(
                            [CalculatedCurrentYearWeightedValue]
                        )
                in
                    if List.Count(Values) = 0 then
                        null
                    else
                        List.Sum(Values),
            type nullable number
        }
    },
    SortedRows =
        Table.Sort(
            GroupedRows,
            {
                {"ForecastReadinessStatus", Order.Ascending},
                {"TotalWeightedEstimatedValue", Order.Descending}
            }
        )
    in
        SortedRows

```

The completed query returns eight Business Unit rows and reconciles to 106 2026-closing opportunities:

- 75 Ready records with EUR 807,900.00 of reconciled current-year value;
- 28 Not Ready records with EUR 1,989,178.99 of weighted pipeline because current-year allocation is missing;
- 2 Not Ready records with EUR 50,000.00 of estimated value because Probability is missing; and
- 1 Review record with EUR 13,500.00 of calculated current-year value because its two-year allocation totals 110%.

Inference – Forecast Readiness Control

The query establishes a clear inclusion boundary for CRM forecast use. Ready records have both the required business inputs and a reconciled current-year calculation. Not Ready and Review records remain visible but are not silently folded into the forecast. This preserves both financial transparency and data- quality accountability.

17.12 Completed Forecast Readiness Status Summary

The Business Unit detail was summarized into one row per forecast-readiness status using:

```
DQ_Opportunity_2026ForecastReadinessStatusSummary
```

The implemented Power Query logic is:

```
let
```

```

Source = DQ_Opportunity_2026ForecastReadinessSummary,

GroupedRows =
    Table.Group(
        Source,
        {
            "ForecastReadinessStatus"
        },
        {
            {
                "RecordCount",
                each List.Sum([RecordCount]),
                Int64.Type
            },
            {
                "TotalEstimatedValue",
                each List.Sum(
                    List.RemoveNulls([TotalEstimatedValue])
                ),
                type nullable number
            },
            {
                "TotalWeightedEstimatedValue",
                each List.Sum(
                    List.RemoveNulls([TotalWeightedEstimatedValue])
                ),
                type nullable number
            },
            {
                "TotalCalculatedCurrentYearValue",
                each
                    let
                        Values =
                            List.RemoveNulls(
                                [TotalCalculatedCurrentYearValue]
                            )
                    in
                        if List.Count(Values) = 0 then
                            null
                        else
                            List.Sum(Values),
                type nullable number
            }
        }
    ),

SortedRows =
    Table.Sort(
        GroupedRows,
        {
            {"RecordCount", Order.Descending}
        }
    )

in
    SortedRows

```

The four status rows reconcile exactly to 106 records. The Ready status contains EUR 807,900.00

of reconciled current-year value. The cross-year allocation review contains a further EUR 13,500.00 of calculated current-year value that is not yet cleared for inclusion.

Inference – Readiness Status Summary Control

The grouped query converts the detailed control into a compact executive view. It makes a crucial distinction between forecast value that is currently usable, value that is calculated but under review, and financially material pipeline that cannot yet be allocated into 2026.

17.13 Completed 2026 CRM Forecast Bridge

The four-row forecast bridge has now been created and assigns an explicit treatment to each readiness status without mixing values that have different financial meanings.

Create a new reference query from `DQ_Opportunity_2026ForecastReadinessStatusSummary` and name it:

```
DQ_Opportunity_2026ForecastBridge
```

In Advanced Editor, use:

```
let
    Source =
        DQ_Opportunity_2026ForecastReadinessStatusSummary,

    AddedForecastTreatment =
        Table.AddColumn(
            Source,
            "ForecastTreatment",
            each
                if [ForecastReadinessStatus] =
                    "Ready - Reconciled Current-Year Value" then
                        "Include - Ready Current-Year Forecast"

                    else if [ForecastReadinessStatus] =
                        "Review - Cross-Year Allocation Above 100%" then
                            "Hold - Current-Year Value Under Review"

                    else if [ForecastReadinessStatus] =
                        "Not Ready - Missing Current-Year Allocation" then
                            "At Risk - Weighted Pipeline Not Allocated to 2026"

                    else
                        "At Risk - Estimated Pipeline Missing Probability",
                type text
        ),

    AddedReadyCurrentYearForecast =
        Table.AddColumn(
            AddedForecastTreatment,
            "ReadyCurrentYearForecastValue",
            each
                if [ForecastReadinessStatus] =
                    "Ready - Reconciled Current-Year Value"
                then [TotalCalculatedCurrentYearValue]
                else null,
            type nullable number
        ),
```

```

AddedReviewCurrentYearValue =
  Table.AddColumn(
    AddedReadyCurrentYearForecast,
    "ReviewCurrentYearValue",
    each
      if [ForecastReadinessStatus] =
        "Review - Cross-Year Allocation Above 100%"
      then [TotalCalculatedCurrentYearValue]
      else null,
    type nullable number
  ),

AddedAtRiskWeightedPipeline =
  Table.AddColumn(
    AddedReviewCurrentYearValue,
    "AtRiskWeightedPipeline",
    each
      if [ForecastReadinessStatus] =
        "Not Ready - Missing Current-Year Allocation"
      then [TotalWeightedEstimatedValue]
      else null,
    type nullable number
  ),

AddedMissingProbabilityEstimatedValue =
  Table.AddColumn(
    AddedAtRiskWeightedPipeline,
    "MissingProbabilityEstimatedValue",
    each
      if [ForecastReadinessStatus] =
        "Not Ready - Missing Probability"
      then [TotalEstimatedValue]
      else null,
    type nullable number
  ),

SelectedColumns =
  Table.SelectColumns(
    AddedMissingProbabilityEstimatedValue,
    {
      "ForecastReadinessStatus",
      "ForecastTreatment",
      "RecordCount",
      "ReadyCurrentYearForecastValue",
      "ReviewCurrentYearValue",
      "AtRiskWeightedPipeline",
      "MissingProbabilityEstimatedValue"
    }
  )

in
  SelectedColumns

```

The expected control values are:

- Ready current-year forecast: EUR 807,900.00;
- current-year value held for review: EUR 13,500.00;

- at-risk weighted pipeline with missing current-year allocation: EUR 1,989,178.99; and
- estimated pipeline with missing Probability: EUR 50,000.00.

These four values remain in separate columns. They are not four additive components of the same accounting measure. The completed bridge therefore communicates forecast treatment clearly before Business Unit harmonization and semantic-model implementation.

Inference – CRM Forecast Bridge Control

The bridge creates a governed boundary between forecast inclusion and unresolved pipeline. It prevents financially different concepts from being collapsed into one number and gives the future semantic model four explicit treatments rather than forcing missing information into zero-valued forecast rows.

17.14 Completed Cross-Source Business Unit Vocabulary Audit

A dedicated control query was created to inventory the Business Unit values used by Fact_ActualSales, Fact_ProjectAccrual and Fact_Opportunities:

```
DQ_BusinessUnit_SourceValues
```

The audit confirms:

- Actual Sales: 346 ExEd, 120 Qualification Programs and 329 University Programs;
- CRM Opportunities: 92 ExEd Programs BU, 7 Qualification Programs BU and 27 University Programs BU;
- Project Accrual: 162 ExEd Programs BU, 43 Qualification Programs BU, 104 University Programs BU and 4 missing Business Unit values.

Inference – Business Unit Source Audit

The three datasets describe the same three populated Business Units, but Actual Sales uses a shorter naming convention. CRM and Accrual already agree. The four missing Accrual Business Units must remain unassigned until validated. An explicit shared dimension is therefore safer and more auditable than replacing labels ad hoc inside each source.

17.15 Completed DimBusinessUnit

Create a new query group named:

```
04_Dimensions
```

Then create a Blank Query named:

```
DimBusinessUnit
```

Use the following Power Query code:

```
let
    Source =
        #table(
            type table [
                BusinessUnitKey = Int64.Type,
                BusinessUnitName = text,
                BusinessUnitShortName = text,
                BusinessUnitSortOrder = Int64.Type
            ]
        )
```

```

    ],
    {
        {
            1,
            "ExEd Programs BU",
            "ExEd",
            1
        },
        {
            2,
            "Qualification Programs BU",
            "Qualification Programs",
            2
        },
        {
            3,
            "University Programs BU",
            "University Programs",
            3
        },
        {
            0,
            "Unassigned / Missing BU",
            "Unassigned",
            99
        }
    }
)

in
Source

```

The completed dimension contains exactly four rows. The fourth row is a technical member for records whose Business Unit is genuinely missing; it does not infer or fabricate a business classification. All four dimension columns are fully populated and the four keys are unique.

Inference – Canonical Business Unit Dimension

The Business Unit vocabulary is now governed independently from the source systems. Keys 1–3 represent the three actual Business Units and key 0 provides a controlled destination for genuinely unassigned records. This design avoids guessing the Business Unit of incomplete accrual rows while enabling consistent cross-source filtering.

17.16 Completed Actual Sales Business Unit Key Mapping

A BusinessUnitKey column was added to Fact_ActualSales. The original BusinessUnitSource was not removed.

The implemented mapping is:

```

if [BusinessUnitSource] = "ExEd" then
    1
else if [BusinessUnitSource] = "Qualification Programs" then
    2
else if [BusinessUnitSource] = "University Programs" then
    3
else
    0

```

The field was typed as `Int64.Type`. The result contains 795 rows, 100% valid keys, 0 errors, 0 empty keys and 3 distinct key values. The distribution is 346 rows on key 1, 120 rows on key 2 and 329 rows on key 3. No Actual Sales row maps to key 0.

Inference – Actual Sales BU-Key Integration

The shorter Actual Sales source labels no longer prevent consistent Business Unit modelling. Every row now carries a stable relationship key, but the original source text remains available for audit and lineage. The mapping also reconciles exactly to the earlier Business Unit source audit.

17.17 Completed Project Accrual Business Unit Key

The `Fact_ProjectAccrual` mapping has now been completed. A Custom Column named `BusinessUnitKey` was added:

```
BusinessUnitKey
```

Use:

```
if [BusinessUnitSource] = null
    or Text.Trim(Text.From([BusinessUnitSource])) = "" then
    0
else if [BusinessUnitSource] = "ExEd Programs BU" then
    1
else if [BusinessUnitSource] = "Qualification Programs BU" then
    2
else if [BusinessUnitSource] = "University Programs BU" then
    3
else
    null
```

Then set the column type to Whole Number / `Int64.Type`.

Based on the completed source audit, the expected distribution is:

- key 1: 162 records;
- key 2: 43 records;
- key 3: 104 records; and
- key 0: 4 records with missing Business Unit.

The completed result remains at 313 rows and the key column is 100% valid, with 0 errors, 0 empty values and 4 distinct keys. The earlier source-value audit reconciles those keys to 162 / 43 / 104 / 4 records on keys 1 / 2 / 3 / 0 respectively. The `else null` branch remains deliberately in the transformation so an unexpected future BU label will surface as an unmapped value rather than being silently assigned.

Inference – Project Accrual BU-Key Integration

Project Accrual is now dimension-ready. The explicit key-0 treatment preserves the four source records with missing Business Unit as an auditable exception while still allowing them to participate in the model under the Unassigned dimension member.

17.18 Completed CRM Opportunity Business Unit Key

The same canonical mapping was then applied to `Fact_Opportunities`:

```

if [BusinessUnitSource] = null
    or Text.Trim(Text.From([BusinessUnitSource])) = "" then
    0
else if [BusinessUnitSource] = "ExEd Programs BU" then
    1
else if [BusinessUnitSource] = "Qualification Programs BU" then
    2
else if [BusinessUnitSource] = "University Programs BU" then
    3
else
    null

```

The key was typed as `Int64.Type`. The completed profile contains 126 rows, 100% valid values, 0 errors, 0 empty values and 3 distinct keys. The distribution reconciles to 92 / 7 / 27 records on keys 1 / 2 / 3, with no records on key 0.

Inference – CRM Opportunity BU-Key Integration

All three analytical facts now carry the same stable Business Unit relationship key. Source-specific Business Unit text remains preserved for lineage, while the semantic model can use one shared dimension without relying on text joins.

17.19 Completed Model-Load Cleanup and Business Unit Relationships

Before relationship validation, loading was disabled for all staging and data-quality queries. They remain in Power Query with their applied steps and continue to support transformation, reconciliation and audit logic.

The visible model now contains only:

- `DimBusinessUnit`;
- `Fact_ActualSales`;
- `Fact_ProjectAccrual`; and
- `Fact_Opportunities`.

Three relationships have been created on `BusinessUnitKey`. In each case, `DimBusinessUnit` is on the one side and the fact table is on the many side. The relationships are active and use Single cross-filter direction. The selected Actual Sales relationship is explicitly shown by Power BI as **Many to one** (*:1) from fact to dimension, which is the same star-schema cardinality viewed from the fact side.

Inference – Business Unit Relationship Model

The Business Unit model is now structurally clean: there are no loaded staging or DQ helper tables and no fact-to-fact Business Unit relationships. One dimension filters all three facts through stable keys. This limits ambiguous filter paths and establishes the correct star-schema direction before time and project dimensions are added.

17.20 Completed Business Unit Filter-Propagation Validation

The three Business Unit relationships have now been validated with row-count measures to confirm that filters propagate exactly as intended.

Create the following three measures:

```
Actual Row Count =
COALESCE(
    COUNTROWS(Fact_ActualSales),
    0
)
```

```
Accrual Row Count =
COALESCE(
    COUNTROWS(Fact_ProjectAccrual),
    0
)
```

```
Opportunity Row Count =
COALESCE(
    COUNTROWS(Fact_Opportunities),
    0
)
```

Create a Table or Matrix visual with `DimBusinessUnit[BusinessUnitName]` on rows and the three measures as values. The expected validation result is:

Business Unit	Actual Rows	Accrual Rows	Opportunity Rows
ExEd Programs BU	346	162	92
Qualification Programs BU	120	43	7
University Programs BU	329	104	27
Unassigned / Missing BU	0	4	0
Total	795	313	126

The completed visual reconciles exactly to the expected values shown above. The totals are 795 Actual rows, 313 Accrual rows and 126 Opportunity rows. The Unassigned member contains 0 Actual rows, 4 Accrual rows and 0 Opportunity rows, which confirms that the four missing-BU accrual records are isolated correctly and no other fact record is misclassified.

Inference – Business Unit Filter Propagation

The row-count test validates the Business Unit relationships functionally, not only structurally. Each Business Unit produces the expected row counts in all three facts, and the grand totals reconcile exactly to the fact-table row counts. This confirms that `DimBusinessUnit` can be used safely as a shared slicer across Actual Sales, Project Accrual and CRM Opportunities.

17.21 Completed 2026 DimDate Creation

A new query named `DimDate` was created with a complete daily calendar from 1 January through 31 December 2026.

The resulting table contains 365 rows and seven columns:

- `Date`;
- `Year`;
- `MonthNumber`;
- `MonthName`;

- YearMonth;
- MonthStart; and
- Quarter.

The profile confirms 365 unique dates, zero errors, zero empty dates, 12 MonthNumber values, 12 MonthName values, 12 YearMonth values, 12 MonthStart values and 4 Quarter values.

The current Power Query layout places DimDate under 01_Staging. The data itself is correct, but the query should be moved to 04_Dimensions before Date relationships are created. This is a query-organization correction only and does not require changing the M code.

Inference – Date Dimension Readiness

The calendar has a unique daily key and complete 2026 coverage, so it is ready to act as the one side of Date relationships. Actual Sales covers January– August through MonthStart, while Project Accrual covers September– December through ForecastMonth; a shared Date dimension will therefore support a continuous full-year 2026 view without appending the two facts.

17.22 Completed DimDate Placement and Date Relationships

DimDate is now in the governed dimension layer. The Date relationships were then created in Model View:

Dimension Side	Fact Side	Cardinality	Filter
DimDate [Date]	Fact_ActualSales [MonthStart]	1:*	Single
DimDate [Date]	Fact_ProjectAccrual [ForecastMonth]		Single

Both relationships are active. Power BI displays them from the fact-table perspective as Many to one (*:1), with DimDate on the one side. Both use Cross-filter direction = Single.

Fact_Opportunities remains intentionally disconnected from DimDate. CRM contains multiple date roles (EstimatedClosingDate, ClosingDate and SimulatedProgramStartDate) and records outside 2026. The CRM Date strategy will therefore be designed separately.

Inference – Date Relationship Structure

The semantic model now has two clean conformed dimensions: DimBusinessUnit and DimDate. Date filters flow from one calendar to Actual Sales and Project Accrual without creating fact-to-fact relationships or bidirectional ambiguity. CRM remains deliberately outside this Date path until its multiple date roles are modelled explicitly.

17.23 Completed Date Filter-Propagation Validation

The existing row-count measures were reused with DimDate[MonthName] to verify the Date relationships operationally.

The completed month-level result is:

Month	Actual Row Count	Accrual Row Count
January	105	0
February	96	0
March	112	0
April	113	0
May	115	0
June	110	0
July	50	0
August	94	0
September	0	87
October	0	76
November	0	84
December	0	66
Total	795	313

The January–August Actual counts sum exactly to 795, while the September–December Accrual counts sum exactly to 313. No Accrual row appears in the January–August period and no Actual row appears in the September–December period.

Inference – Date Filter Propagation

The Date relationships are now validated functionally, not only visually. The shared calendar reproduces the known source horizons exactly: Actual Sales supplies the first eight months of 2026 and Project Accrual supplies the last four months. This proves that a single Date axis can support a continuous full-year 2026 view without appending the two facts or introducing double-counting through the relationship layer.

17.24 Completed Month Sorting and Base Financial Measures

The month-order control has now been completed. `DimDate[MonthName]` is displayed chronologically from January through December using `MonthNumber` as its sort column.

The first three financial measures have also been created:

```
Actual Net Sales =
SUM(Fact_ActualSales[ActualNetSales])
```

```
Project Accrual Forecast =
SUM(Fact_ProjectAccrual[ProjectAccrualForecast])
```

```
Base Full-Year Forecast =
COALESCE([Actual Net Sales], 0)
+ COALESCE([Project Accrual Forecast], 0)
```

The completed month-level visual shows:

Month	Actual Net Sales	Project Accrual Forecast	Base Full-Year Forecast
January	EUR 1,727,161.80	–	EUR 1,727,161.80
February	EUR 1,682,476.45	–	EUR 1,682,476.45
March	EUR 1,977,142.41	–	EUR 1,977,142.41
April	EUR 2,514,361.93	–	EUR 2,514,361.93
May	EUR 2,710,177.12	–	EUR 2,710,177.12
June	EUR 1,736,234.71	–	EUR 1,736,234.71
July	EUR 429,048.56	–	EUR 429,048.56
August	EUR 1,858,464.84	–	EUR 1,858,464.84
September	–	EUR 1,866,125.46	EUR 1,866,125.46
October	–	EUR 1,706,279.10	EUR 1,706,279.10
November	–	EUR 1,807,452.49	EUR 1,807,452.49
December	–	EUR 1,236,168.73	EUR 1,236,168.73
Total	EUR 14,635,067.81	EUR 6,616,025.78	EUR 21,251,093.59

The grand totals reconcile exactly to the source-level controls already documented in the report:

```

EUR 14,635,067.81
+ EUR 6,616,025.78
= EUR 21,251,093.59

```

The displayed monthly values are formatted to two decimals independently. Accordingly, a one-cent difference can arise when manually summing rounded monthly display values versus evaluating the DAX grand total over the underlying source precision. The authoritative control is the Power BI grand total, which matches the previously reconciled source totals.

Source-Aligned Accrual Basis

The EUR 6,616,025.78 Project Accrual Forecast remains the source-aligned amount. The earlier duplicate analysis identified EUR 175,318.63 of potential duplicate overstatement, but no accrual rows have been removed because Finance validation is still pending. Therefore, EUR 21,251,093.59 is the controlled *source-aligned base forecast*, not a Finance-approved duplicate-adjusted forecast.

Inference – Base Full-Year Forecast

The first two financial layers are now operational on one shared 2026 calendar. Actual Net Sales supplies the realized January–August value and Project Accrual supplies the September–December ongoing-project forecast. The two measures combine into a continuous 12-month base forecast of EUR 21,251,093.59 before CRM opportunity contribution is added. This validates both the relationship design and the financial aggregation logic at the annual and monthly levels.

17.25 Completed Business Unit Financial Validation

The three base financial measures were validated through `DimBusinessUnit[BusinessUnitName]`. The completed result is:

Business Unit	Actual Net Sales	Project Accrual Forecast	Base Full-Year Forecast
ExEd Programs BU	EUR 6,044,545.34	EUR 3,009,546.88	EUR 9,054,092.22
Qualification Programs BU	EUR 4,866,104.98	EUR 1,772,854.87	EUR 6,638,959.85
University Programs BU	EUR 3,724,417.49	EUR 1,791,074.03	EUR 5,515,491.52
Unassigned / Missing BU	EUR 0.00	EUR 42,550.00	EUR 42,550.00
Total	EUR 14,635,067.81	EUR 6,616,025.78	EUR 21,251,093.59

The Business Unit rows reconcile exactly to the previously validated annual totals. The explicit Unassigned member contains no Actual Net Sales and exactly EUR 42,550.00 of Project Accrual Forecast, matching the four known missing-BU accrual records.

Inference – Business Unit Financial Validation

The Business Unit relationship layer is now validated for both record counts and financial values. The same dimension that reconciled 795 Actual, 313 Accrual and 126 CRM rows also propagates the financial measures without loss or misallocation. Keeping EUR 42,550.00 in the explicit Unassigned member preserves the source exception instead of forcing it into a real Business Unit.

17.26 Completed Date-Table Designation

DimDate has now been formally marked as the Power BI model Date table using:

```
Table tools
-> Mark as date table
-> Date column: DimDate[Date]
```

This completes the model metadata control for the shared 2026 calendar.

Inference – Date-Table Designation

The calendar is no longer only a technically valid dimension with active relationships; it is also explicitly designated as the model Date table. This provides a clear semantic time backbone for current and future time-intelligence measures while preserving the deliberate decision not to force CRM onto a single Date role yet.

17.27 Completed Ready CRM Measures and Forecast Inclusion

Two governed CRM measures were created to expose only the technically cleared 2026 current-year CRM contribution.

The Ready opportunity count is:

```
CRM Ready Opportunity Count =
CALCULATE(
    COUNTROWS(Fact_Opportunities),
    FILTER(
        Fact_Opportunities,
        YEAR(Fact_Opportunities[ClosingDate]) = 2026
        && Fact_Opportunities[CurrentYearWeightedCheck] = "Match"
        && Fact_Opportunities[OpportunityQualityStatus]
```

```

        <> "Critical - Allocation Above 100%"
    )
)

```

The Ready current-year value is:

```

CRM Ready Current-Year Value =
CALCULATE(
    SUM(Fact_Opportunities[CurrentYearWeightedEstimatedValue]),
    FILTER(
        Fact_Opportunities,
        YEAR(Fact_Opportunities[ClosingDate]) = 2026
        && Fact_Opportunities[CurrentYearWeightedCheck] = "Match"
        && Fact_Opportunities[OpportunityQualityStatus]
            <> "Critical - Allocation Above 100%"
    )
)

```

The completed Business Unit validation is:

Business Unit	Ready Opportunity Count	Ready Current-Year Value
ExEd Programs BU	60	EUR 527,877.50
Qualification Programs BU	2	EUR 35,250.00
University Programs BU	13	EUR 244,772.50
Unassigned / Missing BU	0	EUR 0.00
Total	75	EUR 807,900.00

The Ready CRM contribution was then added to the source-aligned base forecast:

```

Forecast Including Ready CRM =
COALESCE([Base Full-Year Forecast], 0)
+ COALESCE([CRM Ready Current-Year Value], 0)

```

The completed Business Unit result is:

Business Unit	Base Full-Year Forecast	Ready CRM Value	Forecast Including Ready CRM
ExEd Programs BU	EUR 9,054,092.22	EUR 527,877.50	EUR 9,581,969.72
Qualification Programs BU	EUR 6,638,959.85	EUR 35,250.00	EUR 6,674,209.85
University Programs BU	EUR 5,515,491.52	EUR 244,772.50	EUR 5,760,264.02
Unassigned / Missing BU	EUR 42,550.00	EUR 0.00	EUR 42,550.00
Total	EUR 21,251,093.59	EUR 807,900.00	EUR 22,058,993.59

Annual / Business Unit CRM Inclusion Only

The Ready CRM measure is intentionally validated at annual and Business Unit level. It must not yet be interpreted as a month-level CRM forecast through DimDate, because Fact_Opportunities has multiple candidate date roles and is deliberately not connected to the 2026 Date dimension at this stage.

Inference – Governed Forecast Including Ready CRM

The source-aligned base forecast of EUR 21,251,093.59 can now be combined with EUR 807,900.00 of technically cleared current-year CRM value without mixing in unresolved pipeline. The current governed annual forecast is therefore EUR 22,058,993.59. The CRM contribution is also fully reconciled by Business Unit: ExEd contributes EUR 527,877.50, Qualification Programs EUR 35,250.00 and University Programs EUR 244,772.50.

17.28 Immediate Next Step – Semantic CRM Governance Measures

The next step is to surface the three remaining CRM forecast-treatment buckets as semantic-model measures while keeping them separate from the Ready forecast.

Create:

```
CRM Review Current-Year Value =
CALCULATE(
    SUM(Fact_Opportunities[CurrentYearWeightedEstimatedValue]),
    FILTER(
        Fact_Opportunities,
        YEAR(Fact_Opportunities[ClosingDate]) = 2026
        && NOT ISBLANK(Fact_Opportunities[Probability])
        && NOT ISBLANK(Fact_Opportunities[CurrentYearAllocationPct])
        && NOT ISBLANK(Fact_Opportunities[NextYearAllocationPct])
        && Fact_Opportunities[CurrentYearAllocationPct]
            + Fact_Opportunities[NextYearAllocationPct] > 100
    )
)
```

Expected result:

EUR 13,500.00

Then create:

```
CRM At-Risk Weighted Pipeline =
CALCULATE(
    SUM(Fact_Opportunities[WeightedEstimatedValue]),
    FILTER(
        Fact_Opportunities,
        YEAR(Fact_Opportunities[ClosingDate]) = 2026
        && NOT ISBLANK(Fact_Opportunities[Probability])
        && ISBLANK(Fact_Opportunities[CurrentYearAllocationPct])
    )
)
```

Expected result:

EUR 1,989,178.99

Then create:

```
CRM Missing Probability Estimated Value =
CALCULATE(
    SUM(Fact_Opportunities[EstimatedValue]),
    FILTER(
        Fact_Opportunities,
        YEAR(Fact_Opportunities[ClosingDate]) = 2026
        && ISBLANK(Fact_Opportunities[Probability])
    )
)
```

Expected result:

EUR 50,000.00

Optionally create a review-approved scenario:

```
Forecast If Review Approved =
[Forecast Including Ready CRM]
+ [CRM Review Current-Year Value]
```

Expected result:

EUR 22,072,493.59

Use separate Card visuals for the governed values. Do not add the At-Risk weighted pipeline or the Missing-Probability exposure into **Forecast Including Ready CRM**; they remain unresolved pipeline, not cleared current-year forecast.

Inference – Why the Risk Buckets Stay Separate

The next semantic layer should communicate forecast confidence rather than collapse all pipeline into one number. EUR 807,900.00 is cleared for inclusion, EUR 13,500.00 is a calculable amount held for review, EUR 1,989,178.99 is probability-weighted pipeline whose 2026 allocation is unknown, and EUR 50,000.00 is estimated pipeline whose Probability is missing. Keeping these buckets separate prevents false precision and creates an interview-ready forecast bridge.

17.29 Completed Semantic CRM Governance Measures and KPI Presentation

The three CRM governance measures documented previously as the immediate next semantic-model control have now been created and validated in the report model. The completed measures are:

```
CRM Review Current-Year Value =
CALCULATE(
    SUM(Fact_Opportunities[CurrentYearWeightedEstimatedValue]),
    FILTER(
        Fact_Opportunities,
        YEAR(Fact_Opportunities[ClosingDate]) = 2026
        && NOT ISBLANK(Fact_Opportunities[Probability])
        && NOT ISBLANK(Fact_Opportunities[CurrentYearAllocationPct])
        && NOT ISBLANK(Fact_Opportunities[NextYearAllocationPct])
        && Fact_Opportunities[CurrentYearAllocationPct]
            + Fact_Opportunities[NextYearAllocationPct] > 100
    )
)
```

```
CRM At-Risk Weighted Pipeline =
CALCULATE(
    SUM(Fact_Opportunities[WeightedEstimatedValue]),
    FILTER(
        Fact_Opportunities,
        YEAR(Fact_Opportunities[ClosingDate]) = 2026
        && NOT ISBLANK(Fact_Opportunities[Probability])
        && ISBLANK(Fact_Opportunities[CurrentYearAllocationPct])
    )
)
```

```
CRM Missing Probability Estimated Value =
CALCULATE(
    SUM(Fact_Opportunities[EstimatedValue]),
    FILTER(
        Fact_Opportunities,
        YEAR(Fact_Opportunities[ClosingDate]) = 2026
        && ISBLANK(Fact_Opportunities[Probability])
    )
)
```

```
Forecast If Review Approved =
[Forecast Including Ready CRM]
+ [CRM Review Current-Year Value]
```

The resulting governed values are:

Semantic Measure	Validated Result
CRM Review Current-Year Value	EUR 13,500.00
CRM At-Risk Weighted Pipeline	EUR 1,989,178.99
CRM Missing Probability Estimated Value	EUR 50,000.00
Forecast If Review Approved	EUR 22,072,493.59

The KPI cards have also been changed from the earlier dollar-symbol display to EUR formatting. The current Card visual still abbreviates large numbers using K and M display units. This is a presentation setting only; it does not change the underlying DAX values shown above.

Inference – Completed CRM Governance Measures

The semantic layer now distinguishes four different confidence states instead of collapsing them into one pipeline number. The EUR 807,900.00 Ready CRM value remains the only CRM amount automatically included in the governed forecast. EUR 13,500.00 is calculable but held for business review, EUR 1,989,178.99 is probability-weighted pipeline that cannot yet be allocated to the current year, and EUR 50,000.00 represents estimated pipeline whose Probability is missing. This prevents unresolved CRM data from being presented as recognized or cleared forecast revenue.

17.30 Completed Disconnected Forecast-Bucket Dimension

A dedicated table named DimForecastBucket is now present in the semantic model. It contains the fields:

- BucketKey;

- ForecastBucket; and
- SortOrder.

The dimension is intentionally disconnected from the fact tables. The forecast-bridge measures use the selected bucket label to present the governed forecast states without adding another filter path between the analytical facts.

The bridge currently exposes the following business treatments:

Forecast Bucket	Amount	Treatment
Base Full-Year Forecast	EUR 21,251,093.59	Included – Core Forecast
Ready CRM – Included	EUR 807,900.00	Included – Ready CRM
Review CRM – Held	EUR 13,500.00	Held – Business Review
At-Risk Pipeline – Not Included	EUR 1,989,178.99	Not Included – Allocation Missing
Missing Probability – Not Included	EUR 50,000.00	Not Included – Probability Missing

Inference – Forecast Bucket Dimension

Keeping DimForecastBucket disconnected is deliberate. It acts as a presentation and scenario-control dimension rather than a business filter on Actual Sales, Accrual or CRM. This avoids ambiguous relationships while giving the dashboard an auditable bridge from the source-aligned base forecast to Ready, Review and unresolved CRM states.

17.31 Completed CRM Calendar Dimension

A separate CRM calendar named DimCRMDate has now been created in the 04_Dimensions layer. It covers the full CRM horizon from 1 January 2024 through 31 December 2028.

The Power Query implementation is:

```
let
    StartDate = #date(2024, 1, 1),
    EndDate = #date(2028, 12, 31),

    DateList =
        List.Dates(
            StartDate,
            Duration.Days(EndDate - StartDate) + 1,
            #duration(1, 0, 0, 0)
        ),

    ToTable =
        Table.FromList(
            DateList,
            Splitter.SplitByNothing(),
            {"Date"}
        ),

    ChangedType =
        Table.TransformColumnTypes(
            ToTable,
            {"Date", type date}
        )
```

```
    ),  
  
    AddedYear =  
        Table.AddColumn(  
            ChangedType,  
            "Year",  
            each Date.Year([Date]),  
            Int64.Type  
        ),  
  
    AddedMonthNumber =  
        Table.AddColumn(  
            AddedYear,  
            "MonthNumber",  
            each Date.Month([Date]),  
            Int64.Type  
        ),  
  
    AddedMonthName =  
        Table.AddColumn(  
            AddedMonthNumber,  
            "MonthName",  
            each Date.MonthName([Date]),  
            type text  
        ),  
  
    AddedYearMonth =  
        Table.AddColumn(  
            AddedMonthName,  
            "YearMonth",  
            each Date.Year([Date]) * 100 + Date.Month([Date]),  
            Int64.Type  
        ),  
  
    AddedMonthStart =  
        Table.AddColumn(  
            AddedYearMonth,  
            "MonthStart",  
            each Date.StartOfMonth([Date]),  
            type date  
        )  
  
in  
    AddedMonthStart
```

Full-dataset profiling confirms:

CRM Date Metric	Result
Calendar period	1 January 2024 – 31 December 2028
Rows	1,827
Date errors	0
Date empty values	0
Distinct / unique Date values	1,827 / 1,827
Distinct Year values	5
Distinct MonthNumber / MonthName values	12 / 12
Distinct YearMonth values	60
Distinct MonthStart values	60

The row count of 1,827 is correct for the inclusive five-year period because both 2024 and 2028 are leap years.

Inference – CRM Calendar Readiness

The CRM time horizon is now covered by a complete, unique and error-free daily calendar that is independent of the 2026 operational calendar used by Actual Sales and Project Accrual. This prevents the multi-year CRM pipeline from being forced into the narrower 2026 Date dimension and creates a clean foundation for explicit CRM date roles.

17.32 Completed CRM Date-Role Relationships

The CRM Date strategy has now been implemented using one active relationship and one inactive role-playing relationship between DimCRMDate and Fact_Opportunities.

The relationship configuration is:

Dimension Side	Fact Side	Cardinality	Active	Filter
DimCRMDate[Date]	Fact_Opportunities[ClosingDate]	Many to one (*)	Yes	Single
DimCRMDate[Date]	Fact_Opportunities[SimulatedProgramStartDate]	Many to one (*)	No	Single

Power BI displays these relationships from the fact-table perspective as **Many to one (*)**. The solid relationship line represents the active **ClosingDate** role; the dotted relationship line represents the inactive **SimulatedProgramStartDate** role. **EstimatedClosingDate** remains unconnected at this stage.

The existing Business Unit relationships remain unchanged: one governed DimBusinessUnit continues to filter all three facts with active, single-direction relationships. The 2026 DimDate relationships to Actual Sales and Project Accrual also remain unchanged.

Inference – CRM Date-Role Model

The relationship structure is correct and avoids competing active date paths. Closing Date provides the default CRM time context, while Program Start Date is available explicitly through DAX when required. This lets the same opportunity be analysed by sales-closing timing or delivery/program-start timing without creating ambiguous bidirectional filters or duplicating the CRM fact table.

17.33 Immediate Next Step – Validate CRM Date Roles with DAX

The next control is to prove that the active and inactive CRM Date relationships behave differently and both reconcile to the same source population.

First create a measure that uses the active **ClosingDate** relationship:

```
CRM Opportunity Count by Closing Date =  
COUNTROWS(Fact_Opportunities)
```

Then create a second measure that explicitly activates the inactive Program Start relationship:

```
CRM Opportunity Count by Program Start =  
CALCULATE(  
    COUNTROWS(Fact_Opportunities),  
    USERELATIONSHIP(  
        Fact_Opportunities[SimulatedProgramStartDate],  
        DimCRMDate[Date]  
    )  
)
```

Create a Table or Matrix with:

- DimCRMDate[Year] as the first validation row field;
- optionally DimCRMDate[YearMonth] for monthly detail;
- CRM Opportunity Count by Closing Date; and
- CRM Opportunity Count by Program Start.

At the grand-total level, both measures should reconcile to all 126 CRM opportunities because both source date fields are populated and the CRM calendar covers the complete 2024–2028 horizon. The year and month distributions should not necessarily be identical, because an opportunity can close in one period and start its simulated program in another.

Do not create a second active relationship between DimCRMDate and Fact_Opportunities. The Program Start relationship should remain inactive and be activated only inside measures that intentionally analyse that date role.

Inference – What This Validation Proves

If both measures reconcile to 126 at total level while producing different time distributions, the role-playing Date design is working as intended. The model can then answer two different business questions—when opportunities are expected to close and when their programs are expected to start—without changing the underlying opportunity rows or creating ambiguous relationship paths.

17.34 Completed CRM Date-Role DAX Validation Results

The planned CRM Date-role control has now been executed in the report view. The validation matrix uses DimCRMDate[Year] together with the two previously defined measures:

```
CRM Opportunity Count by Closing Date
CRM Opportunity Count by Program Start
```

The completed result is:

Year	CRM Opportunity Count by Closing Date	CRM Opportunity Count by Program Start
2024	1	1
2025	1	1
2026	106	56
2027	15	64
2028	3	4
Total	126	126

The two measures reconcile exactly to all 126 CRM opportunities at grand-total level. Their yearly distributions are intentionally different. In particular, 106 opportunities close in 2026 while only 56 have simulated program start in 2026; in 2027 the pattern reverses to 15 closings and 64 program starts.

CRM Date-Role Validation Result

The active Closing Date relationship and the inactive Program Start relationship both resolve the complete CRM population. The inactive relationship is therefore working correctly through USERELATIONSHIP; it is not a broken or orphaned path.

Inference – CRM Date-Role Validation

The result proves that Closing Date and Program Start represent genuinely different business timelines rather than duplicate date fields. Using one active and one inactive relationship preserves both interpretations without introducing two competing active filter paths. The CRM temporal model is therefore functionally validated, not merely structurally configured.

17.35 Completed Project Dimension Construction and Profiling

A Project dimension has now been created by combining the project identifiers from Actual Sales and Project Accrual. CRM Opportunities are not included in this dimension because the supplied CRM source contains an opportunity description rather than a confirmed ProjectID mapping; no project assignment is invented without source evidence.

The implemented Power Query logic is:

```
let
    ActualProjects =
        Table.SelectColumns(
            Fact_ActualSales,
            {"ProjectID"}
        ),

    AccrualProjects =
```

```
Table.SelectColumns(
    Fact_ProjectAccrual,
    {"ProjectID"}
),

CombinedProjects =
Table.Combine(
    {
        ActualProjects,
        AccrualProjects
    }
),

CleanedProjects =
Table.TransformColumns(
    CombinedProjects,
    {
        {
            "ProjectID",
            each
                if _ = null then
                    null
                else
                    Text.Trim(Text.Clean(Text.From(_))),
            type text
        }
    }
),

RemovedBlankProjects =
Table.SelectRows(
    CleanedProjects,
    each
        [ProjectID] <> null
        and [ProjectID] <> ""
),

DistinctProjects =
Table.Distinct(
    RemovedBlankProjects,
    {"ProjectID"}
),

SortedProjects =
Table.Sort(
    DistinctProjects,
    {
        {"ProjectID", Order.Ascending}
    }
),

AddedProjectKey =
Table.AddIndexColumn(
    SortedProjects,
    "ProjectKey",
    1,
    1,
    Int64.Type
),
```

```

ReorderedColumns =
    Table.ReorderColumns(
        AddedProjectKey,
        {
            "ProjectKey",
            "ProjectID"
        }
    )

in
    ReorderedColumns

```

The completed Power Query profile shows:

Project-Dimension Metric	Result
Rows	252
ProjectID valid values	100%
ProjectID errors	0
ProjectID empty values	0
Distinct ProjectID values	252
Unique ProjectID values	252
ProjectKey range	1–252

The dimension was subsequently moved from **01_Staging** into the governed **04_Dimensions** query group with model load retained. This completes the query-governance placement that was still pending at the earlier checkpoint.

Inference – Project Dimension Readiness

The combined Actual/Accrual project vocabulary contains 252 clean and unique ProjectIDs, so the dimension provides a valid one-side business key. The later semantic implementation confirmed that a separate fact-table surrogate-key merge was not necessary for this case: both project facts already contain clean **ProjectID** values that resolve to the unique dimension. The table has now become a conformed Project dimension and its relationships have been validated.

17.36 Immediate Next Step – Govern Project Keys and Relationships

Implementation outcome note. This subsection is retained as the original planned control for audit history. During implementation, the proposed Left Outer merge of `ProjectKey` into both facts was found to be unnecessary for the current case because `DimProject[ProjectID]` is unique and both facts already contain clean matching `ProjectID` values. The final implemented relationships therefore use `ProjectID` directly. The completed state is documented in Section 17.37.

The next technical control is not to create another dimension. It is to finish the semantic implementation of the Project dimension already created.

First, move `DimProject` from `01_Staging` to `04_Dimensions`. Keep model load enabled.

Then add `ProjectKey` to both project-based facts using a Left Outer merge on `ProjectID`:

1. merge `Fact_ActualSales[ProjectID]` with `DimProject[ProjectID]` and expand only `ProjectKey`;
2. confirm all 795 Actual Sales rows receive a non-null `ProjectKey`;
3. merge `Fact_ProjectAccrual[ProjectID]` with `DimProject[ProjectID]` and expand only `ProjectKey`;
4. confirm all 313 Accrual rows receive a non-null `ProjectKey`; and
5. apply the changes and create the semantic relationships.

The target relationships are:

Dimension Side	Fact Side	Cardinality	Filter
<code>DimProject[ProjectKey]</code>	<code>Fact_ActualSales[ProjectKey]</code>	1:*	Single
<code>DimProject[ProjectKey]</code>	<code>Fact_ProjectAccrual[ProjectKey]</code>	1:*	Single

Both relationships should be active. No Project relationship should be created to `Fact_Opportunities` unless a real CRM-to-Project identifier becomes available from the source or business owner.

Inference – Why ProjectKey Mapping Is the Next Control

Creating a unique Project dimension is only the structural half of the task. Propagating the surrogate key into the two project-based facts converts the clean project vocabulary into a governed star-schema filter path. Validating 795 Actual rows and 313 Accrual rows after the merge will prove that no project is lost during key standardization. Excluding CRM from this relationship avoids fabricating a project assignment that the supplied source data does not prove.

17.37 Completed Project Relationships and Functional Validation

The Project semantic layer has now been completed. Two relationships were created using the validated natural project identifier rather than introducing an unnecessary additional merge step:

Dimension Side	Fact Side	Cardinality	Filter
<code>DimProject[ProjectID]</code>	<code>Fact_ActualSales[ProjectID]</code>	1:*	Single
<code>DimProject[ProjectID]</code>	<code>Fact_ProjectAccrual[ProjectID]</code>	1:*	Single

Both relationships are active. In the Power BI relationship Properties pane the same structure

is displayed from the fact perspective as **Many to one (*:1)**. No Project relationship has been created to **Fact_Opportunities**, because the supplied CRM data does not provide a confirmed ProjectID that would justify such a mapping.

The functional validation table uses **DimProject[ProjectID]** together with existing fact measures. The completed grand totals are:

Project-Relationship Validation Metric	Result
Actual Row Count	795
Accrual Row Count	313
Actual Net Sales	EUR 14,635,067.81
Project Accrual Forecast	EUR 6,616,025.78
Base Full-Year Forecast	EUR 21,251,093.59

The Project-level rows also demonstrate combined fact filtering. For example, project 07901009 shows EUR 439,132.89 of Actual Net Sales and EUR 308,913.00 of Project Accrual Forecast, producing EUR 748,045.89 of Base Full-Year Forecast for the same ProjectID.

Project Relationship Validation Result

The Project dimension preserves all project-based source records and financial values. The row counts remain 795 Actual and 313 Accrual, and the financial totals remain identical to the previously validated base forecast. This proves that the new Project dimension filters both facts correctly without losing or duplicating value.

Inference – Project Semantic Layer

The shared Project dimension is now functionally complete. A user can select one ProjectID and analyze recognized Actual revenue and remaining Project Accrual forecast through the same governed project filter. Because the totals reconcile exactly before and after introducing the dimension, the relationship layer adds analytical capability without changing the financial baseline. The model now has validated shared dimensions for Business Unit, calendar time and Project, while CRM retains its separate governed date-role and opportunity logic.

17.38 Immediate Next Step – Project Filter Test and Dashboard Refinement

Before final dashboard styling, perform one final interactive Project filter test:

1. add a slicer using **DimProject[ProjectID]**;
2. select 07901009;
3. confirm **Actual Net Sales** = EUR 439,132.89;
4. confirm **Project Accrual Forecast** = EUR 308,913.00;
5. confirm **Base Full-Year Forecast** = EUR 748,045.89; and
6. clear the slicer and confirm the grand total returns to EUR 21,251,093.59.

After this final interaction test, the immediate presentation step is to create a Project forecast-ranking visual for the executive dashboard: rank the top projects by **Base Full-Year Forecast** and expose Actual versus remaining Accrual in tooltips or a companion table.

Inference – Transition to Dashboard Phase

The remaining work is now mainly presentation and communication rather than core data modelling. A Project slicer test closes the last shared-dimension functional control; the model can then be presented through executive KPIs, Business Unit and Project rankings, monthly forecast views and clearly separated CRM risk buckets.

17.39 Completed Interactive Project Filter Validation

The final interactive Project-dimension control has now been executed in Report View using a slicer based on DimProject[ProjectID]. Project 07901009 was selected as the test case.

The report returned:

Project Filter Validation Metric	Result
Selected ProjectID	07901009
Actual Row Count	6
Accrual Row Count	4
Actual Net Sales	EUR 439,132.89
Project Accrual Forecast	EUR 308,913.00
Base Full-Year Forecast	EUR 748,045.89

The arithmetic reconciles exactly:

$$\text{EUR 439,132.89} + \text{EUR 308,913.00} = \text{EUR 748,045.89}$$

The same interaction also confirms an intentional modelling boundary: Project selection affects Actual Sales, Project Accrual and the Base Full-Year Forecast, but it does not create a fabricated filter path into CRM Opportunities. CRM forecast-governance values therefore remain independent unless a real source mapping between opportunities and ProjectID becomes available.

Project Filter Validation Result

The Project slicer now behaves as intended across both project-based facts. The selected ProjectID reconciles row counts and financial values exactly, proving that the two active Project relationships are operational in Report View and not merely structurally present in Model View.

Inference – Interactive Project Filtering

The Project dimension has passed both aggregate and interactive validation. A single business ProjectID can now be used to analyze realized revenue and remaining accrual forecast together, while CRM remains governed by its own opportunity logic. This preserves semantic integrity and avoids implying a Project-to-CRM relationship that is not present in the supplied source data.

17.40 Completed Top-10 Project Forecast Ranking

The first Project-level executive visual has now been created as a horizontal bar chart titled:

Base Full-Year Forecast by ProjectID

The visual uses:

- category: DimProject[ProjectID];
- value: Base Full-Year Forecast;
- visual-level Project filter: Top 10 by Base Full-Year Forecast; and
- descending value order.

The current ranking displayed in the report is:

1. D8001031;
2. L9109051;
3. D8001028;
4. D8001021;
5. 07901009;
6. D8001036;
7. D8001062;
8. 07401158;
9. D8001035; and
10. P8404024.

The ranking uses the same validated Base Full-Year Forecast measure that reconciles globally to EUR 21,251,093.59. No new financial calculation was introduced solely for the chart.

Top-10 Project Ranking Result

The report can now identify the largest project-level contributors to the source-aligned annual forecast. The visual is driven by the governed Project dimension and existing financial measures, so project prioritization remains consistent with the validated semantic model.

Inference – Project Forecast Concentration

Project-level ranking adds management value without changing the forecast baseline. It turns the Project dimension from a navigation control into a prioritization tool: users can immediately identify the projects with the largest full-year financial contribution and then drill into their Actual and Accrual components.

17.41 Immediate Next Step – Top-10 Actual versus Accrual Composition

The next dashboard control is to explain *why* each Top-10 project is large by separating recognized Actual Net Sales from the remaining Project Accrual Forecast.

Create a second horizontal **Stacked bar chart**. Use:

- Y-axis: DimProject[ProjectID];
- X-axis / Values: Actual Net Sales;
- X-axis / Values: Project Accrual Forecast;
- visual filter on ProjectID: Top 10;
- By value: Base Full-Year Forecast; and
- keep the same Project slicer interaction as the existing Project visuals.

A suggested title is:

Top 10 Projects - Actual vs Remaining Accrual

If the stacked chart does not retain the preferred descending order after the second measure is added, keep the existing ranking chart as the authoritative ordering visual and use the stacked chart as the companion composition view. Do not change the underlying measures merely to force presentation order.

Inference – Why Composition Is the Next Visual

The ranking answers which projects are financially largest; the composition view answers whether each project is already largely realized as Actual revenue or still depends on future Accrual. Together the two visuals communicate both scale and forecast maturity without introducing a new business assumption.

17.42 Completed Top-10 Actual versus Accrual Composition

The companion Project composition visual planned in Section 17.41 has now been implemented as a horizontal stacked bar chart titled:

Top 10 Projects - Actual vs Remaining Accrual

The visual uses:

- Y-axis: DimProject[ProjectID];
- values: Actual Net Sales and Project Accrual Forecast;
- Top-10 filtering by Base Full-Year Forecast; and
- the same governed Project relationship path already validated for the ranking chart and Project slicer.

No new financial measure was introduced. The visual decomposes the same source-aligned Base Full-Year Forecast into its realized Actual component and its remaining Accrual component.

Project Composition Visual Result

The report now contains both a Top-10 Project ranking and a companion composition view. The first visual identifies the largest project-level forecast contributors; the second explains how much of each project's annual value has already been recognized as Actual Net Sales and how much still depends on remaining Accrual.

Inference – Project Forecast Maturity

Project size and project maturity can now be interpreted together. A large bar in the ranking view does not by itself reveal whether the project is already realized or still forecast-dependent; the composition chart adds that distinction without changing the validated financial baseline.

17.43 Executive Dashboard KPI and Business Unit Control Layer

A dedicated report page named **Executive Dashboard** has now been created. The first dashboard layer contains eight KPI cards:

Executive KPI	Displayed Value
Actual Net Sales	EUR 14.64M
Project Accrual Forecast	EUR 6.62M
Base Full-Year Forecast	EUR 21.25M
Forecast Including Ready CRM	EUR 22.06M
CRM Ready Current-Year Value	EUR 807.90K
CRM Review Current-Year Value	EUR 13.50K
CRM At-Risk Weighted Pipeline	EUR 1.99M
CRM Missing Probability Estimated Value	EUR 50.00K

All eight cards now use EUR formatting consistently. A report slicer based on:

```
DimBusinessUnit[BusinessUnitName]
```

has also been added and exposes the four governed members:

- ExEd Programs BU;
- Qualification Programs BU;
- University Programs BU; and
- Unassigned / Missing BU.

The existing semantic model already places **DimBusinessUnit** on the one side of active Single-

direction relationships to all three analytical facts, so this slicer uses the validated conformed-dimension path rather than source-specific Business Unit text fields.

Inference – Executive Control Layer

The Executive Dashboard now starts from governed totals rather than raw columns. The KPI row separates the committed base forecast, cleared Ready CRM contribution and unresolved CRM risk categories, while the shared Business Unit slicer provides one consistent management filter across Actual Sales, Project Accrual and CRM.

17.44 Executive Monthly Forecast Visual – Presentation Correction

A 12-month visual titled:

Monthly Base Forecast - Actual vs Accrual

has been added with chronological January–December ordering. The underlying fields are correct:

- X-axis: `DimDate[MonthName]`;
- value: `Actual Net Sales`;
- value: `Project Accrual Forecast`; and
- month order governed by `DimDate[MonthNumber]`.

The visual correctly separates the source periods: Actual Net Sales is present in January–August and Project Accrual Forecast in September–December.

However, the current screenshot shows a percentage y-axis from 0% to 100%. This identifies the selected visual as a **100% stacked column chart**. That chart type normalizes every populated month to 100% and therefore hides the financial difference between, for example, a EUR 0.4M month and a EUR 2.7M month.

The chart must therefore be changed to the standard **Stacked column chart**. No field or measure should be changed.

Do not keep the 100% stacked version for the final dashboard.

The business question is monthly forecast magnitude, not percentage composition. The corrected chart should display a monetary y-axis and preserve the same January–December Actual/Accrual split.

Inference – Monthly Forecast Presentation

The Date model and financial measures are functioning correctly; the remaining issue is visual encoding rather than data logic. Replacing the 100% stacked chart with a standard stacked column chart will make monthly magnitude immediately comparable while retaining the validated Actual-versus-Accrual structure.

17.45 Immediate Next Step – Correct Monthly Chart and Add BU Forecast View

The immediate next action is:

1. select the current monthly chart on the **Executive Dashboard**;
2. in Visualizations, choose **Stacked column chart**, not **100% stacked column chart**;
3. keep `DimDate[MonthName]` on the X-axis;

4. keep Actual Net Sales and Project Accrual Forecast as values;
5. verify that the y-axis changes from percentages to monetary values;
6. select each Business Unit in the slicer and confirm the KPI cards and monthly chart respond consistently; and
7. after that validation, add a Business Unit financial-composition visual using DimBusinessUnit [BusinessUnit] with Actual Net Sales, Project Accrual Forecast and Base Full-Year Forecast.

Inference – Next Dashboard Control

This sequence closes the only visible issue in the current Executive Dashboard and then proves the shared Business Unit dimension at presentation level. Once that interaction is validated, the remaining work is primarily final layout, titles, conditional emphasis and interview storytelling rather than core model construction.

17.46 Completed Executive Monthly Forecast Correction

The presentation correction planned in Sections 17.44–17.45 has now been completed. The visual titled:

Monthly Base Forecast - Actual vs Accrual

now uses a standard stacked column chart rather than the earlier 100% stacked version. The field configuration remains unchanged:

- X-axis: DimDate [MonthName];
- Y-axis value: Actual Net Sales;
- Y-axis value: Project Accrual Forecast; and
- tooltip: Base Full-Year Forecast.

The corrected chart now displays monetary magnitude on the y-axis instead of percentage share. January–August continues to display Actual Net Sales and September–December continues to display Project Accrual Forecast. The month order remains chronological.

The explicit Unassigned / Missing BU test also produces only September–December Accrual bars, which is consistent with the known EUR 42,550.00 unassigned accrual exposure and the absence of Actual Sales for that member.

Monthly Forecast Visual Correction Result

The monthly executive visual now communicates both timing and financial magnitude correctly. No DAX measure, Date relationship or source transformation needed to be changed; only the chart encoding was corrected.

Inference – Correct Monthly Magnitude

The successful correction confirms that the earlier issue was purely a presentation-layer problem. The same validated Date model and measures now produce an executive visual in which month-to-month EUR differences are directly comparable without percentage normalization.

17.47 Completed Executive Business Unit Slicer Validation

The **Executive Dashboard** Business Unit slicer was tested interactively against all four governed members. The test confirms that the same selection propagates to Actual Sales, Project Accrual and CRM governance measures.

The core financial result by selected Business Unit is:

Business Unit	Actual	Accrual	Base Forecast	Forecast + Ready CRM
ExEd Programs BU	EUR 6,044,545.34	EUR 3,009,546.88	EUR 9,054,092.22	EUR 9,581,969.72
Qualification Programs BU	EUR 4,866,104.98	EUR 1,772,854.87	EUR 6,638,959.85	EUR 6,674,209.85
University Programs BU	EUR 3,724,417.49	EUR 1,791,074.03	EUR 5,515,491.52	EUR 5,760,264.02
Unassigned / Missing BU	EUR 0.00	EUR 42,550.00	EUR 42,550.00	EUR 42,550.00

The CRM governance measures under the same slicer context are:

Business Unit	Ready CRM	Review	At-Risk	Missing Probability
ExEd Programs BU	EUR 527,877.50	EUR 0.00	EUR 1,229,613.75	EUR 0.00
Qualification Programs BU	EUR 35,250.00	EUR 0.00	EUR 615,625.00	EUR 0.00
University Programs BU	EUR 244,772.50	EUR 13,500.00	EUR 143,940.24	EUR 50,000.00
Unassigned / Missing BU	EUR 0.00	EUR 0.00	EUR 0.00	EUR 0.00

The card visuals currently render some zero-row filtered results as -. This is an expected DAX BLANK() display when the selected Business Unit has no record in that CRM treatment bucket. It is not evidence of a broken relationship. If a later presentation preference requires explicit EUR 0.00 cards, display-specific COALESCE measures can be added without changing the validated business logic.

Business Unit Slicer Validation Result

All four selections produce the expected financial and CRM distributions. ExEd, Qualification and University each return their previously reconciled Actual, Accrual, Ready CRM and At-Risk values. The Unassigned member returns only EUR 42,550.00 of Project Accrual Forecast and no Actual or CRM contribution, exactly matching the known four missing-BU accrual records.

Inference – Report-Level Relationship Validation

The Business Unit model is now validated not only through row-count matrices and standalone financial tables, but also through live Executive Dashboard interaction. One dimension selection filters all three analytical facts in a business-consistent way, while the explicit Unassigned member prevents missing source classifications from leaking into a real Business Unit.

17.48 Immediate Next Step – Business Unit Forecast Composition

The next executive visual should compare the composition of the base forecast across Business Units without creating any new measure.

Create a standard horizontal **Stacked bar chart** with:

- Y-axis: `DimBusinessUnit[BusinessUnitName]`;
- X-axis / Values: `Actual Net Sales`;
- X-axis / Values: `Project Accrual Forecast`;
- Tooltip: `Base Full-Year Forecast`;
- Tooltip: `Forecast Including Ready CRM`; and
- Tooltip: `CRM Ready Current-Year Value`.

Sort the visual by:

`Base Full-Year Forecast`

in descending order. A suggested title is:

`Base Forecast by Business Unit - Actual vs Remaining Accrual`

With the Business Unit slicer cleared, the expected Base Full-Year Forecast ordering is:

1. ExEd Programs BU – EUR 9,054,092.22;
2. Qualification Programs BU – EUR 6,638,959.85;
3. University Programs BU – EUR 5,515,491.52; and
4. Unassigned / Missing BU – EUR 42,550.00.

Keep the existing measures unchanged. The purpose of this visual is presentation, not a new calculation.

Inference – Why Business Unit Composition Is Next

The dashboard can already answer the annual total, monthly timing and Project concentration questions. The Business Unit composition chart adds the management question of where the source-aligned forecast sits organizationally and whether each Business Unit is already realized as Actual revenue or still depends on remaining Accrual. It reuses the same validated semantic model and therefore adds explanation without adding forecast assumptions.

17.49 Completed Business Unit Forecast Composition

The Business Unit composition visual planned in Section 17.48 has now been implemented on the **Executive Dashboard**. The visual is a standard horizontal stacked bar chart with:

- Y-axis: `DimBusinessUnit[BusinessUnitName]`;

- X-axis / Values: Actual Net Sales;
- X-axis / Values: Project Accrual Forecast; and
- the existing base-forecast measures available for context and tooltip use.

With the Business Unit slicer cleared, the chart displays the expected descending organizational pattern: ExEd Programs BU first, followed by Qualification Programs BU, University Programs BU and the explicit **Unassigned / Missing** BU member. The stacked composition also preserves the distinction between recognized Actual revenue and remaining Accrual forecast.

Business Unit Composition Result

The Executive Dashboard can now answer not only the annual total and monthly timing questions, but also where the base forecast sits organizationally and how much of each Business Unit's annual value has already been realized versus remaining in Accrual.

Inference – Business Unit Forecast Composition

The shared Business Unit dimension is now demonstrated at presentation level as well as model level. The same governed dimension that reconciles row counts and KPI cards also produces a coherent financial composition across all four Business Unit members, including the explicit unassigned accrual exposure.

17.50 Completed Forecast Governance Table

A compact executive control table titled:

Forecast Governance - Included vs At Risk

has now been added to the Executive Dashboard. It uses:

- DimForecastBucket[ForecastBucket];
- measure Forecast Bucket Amount; and
- measure Forecast Bucket Treatment.

The rendered order is the intended governance order and the displayed values are:

Forecast Bucket	Amount	Treatment
Base Full-Year Forecast	EUR 21,251,093.59	Included – Core Forecast
Ready CRM – Included	EUR 807,900.00	Included – Ready CRM
Review CRM – Held	EUR 13,500.00	Held – Business Review
At-Risk Pipeline – Not Included	EUR 1,989,178.99	Not Included – Allocation Missing
Missing Probability – Not Included	EUR 50,000.00	Not Included – Probability Missing

The table therefore keeps the cleared forecast, held amount and unresolved pipeline visibly separate instead of presenting all CRM value as one additive forecast.

Forecast Governance Result

The executive page now contains a direct control explaining which values are included, which are held for business review and which remain outside the governed forecast. The financial meaning of each CRM treatment is visible beside its amount.

Inference – Included versus At-Risk Forecast

The dashboard now communicates forecast confidence as well as forecast magnitude. EUR 21.251M forms the source-aligned core forecast and EUR 807.9K of Ready CRM is cleared for inclusion, while EUR 13.5K remains held and the EUR 1.989M allocation- missing pipeline plus EUR 50K missing-Probability exposure remain explicitly outside the committed forecast.

17.51 Executive Dashboard Analytical Composition Checkpoint

The Executive Dashboard now contains the following complementary management views:

- eight annual financial and CRM-governance KPI cards;
- one governed Business Unit slicer;
- the **Forecast Governance – Included vs At Risk** control table;
- Business Unit Actual-versus-Accrual forecast composition;
- monthly Actual-versus-Accrual timing across January–December; and
- Top-10 Project Actual-versus-remaining-Accrual composition.

This page now covers five distinct executive questions: total forecast magnitude, CRM readiness and risk, organizational concentration, monthly timing and Project concentration. No new source transformations or forecast assumptions were required to create the two latest visuals.

Inference – Executive Dashboard Coverage

The report has moved from model construction into final validation and presentation. The remaining work is no longer core semantic-model development; it is primarily interaction verification, field cleanup, layout refinement and preparation of a concise interview narrative explaining the governed forecast.

17.52 Immediate Next Step – Final Interaction Validation

Because the Business Unit composition and Forecast Governance visuals were added after the earlier slicer test, the next control is to re-run the Business Unit interaction test across the complete Executive Dashboard.

Start with **ExEd Programs BU**. The expected KPI context is:

- Actual Net Sales: EUR 6,044,545.34;
- Project Accrual Forecast: EUR 3,009,546.88;
- Base Full-Year Forecast: EUR 9,054,092.22;
- Forecast Including Ready CRM: EUR 9,581,969.72;
- CRM Ready Current-Year Value: EUR 527,877.50;
- CRM Review Current-Year Value: EUR 0.00 / blank display;

- CRM At-Risk Weighted Pipeline: EUR 1,229,613.75; and
- CRM Missing Probability Estimated Value: EUR 0.00 / blank display.

During this selection, verify that:

1. all eight KPI cards respond;
2. the Forecast Governance table recalculates its bucket amounts;
3. the Business Unit composition chart reflects the selected Business Unit;
4. the monthly chart recalculates the selected Business Unit's monthly profile;
5. the Top-10 Project chart re-ranks projects inside the selected Business Unit;
6. clearing the slicer restores the full-company totals; and
7. no visual remains unintentionally disconnected because of an Edit Interactions setting.

Repeat the same interaction check for Qualification Programs BU, University Programs BU and Unassigned / Missing BU. After this control passes, the next work package is final semantic-field hiding and visual polish rather than another major modelling step.

Inference – Why Final Interaction Validation Is Next

The new visuals reuse already validated measures, but a dashboard is only fully validated when filter propagation is confirmed at report level. Re-testing all four Business Unit selections proves that the complete executive page behaves as one coherent analytical product rather than as a collection of individually correct visuals.

17.53 Completed Business Unit Financial QA

The Business Unit financial QA control has now been completed after the full Executive Dashboard interaction pass. The two reconciliation measures were evaluated at the all-company level and again under each governed Business Unit selection:

```
Base Forecast Reconciliation Variance =  
  [Base Full-Year Forecast]  
  - (  
    [Actual Net Sales]  
    + [Project Accrual Forecast]  
  )
```

```
Ready CRM Inclusion Variance =  
  [Forecast Including Ready CRM]  
  - (  
    [Base Full-Year Forecast]  
    + [CRM Ready Current-Year Value]  
  )
```

The validation states were:

Business Unit Context	Base Forecast	Ready CRM	Reconciliation Variances
All Business Units	EUR 21,251,093.59	EUR 807,900.00	EUR 0.00 / EUR 0.00
ExEd Programs BU	EUR 9,054,092.22	EUR 527,877.50	EUR 0.00 / EUR 0.00
Qualification Programs BU	EUR 6,638,959.85	EUR 35,250.00	EUR 0.00 / EUR 0.00
University Programs BU	EUR 5,515,491.52	EUR 244,772.50	EUR 0.00 / EUR 0.00
Unassigned / Missing BU	EUR 42,550.00	–	EUR 0.00 / EUR 0.00

For the explicit Unassigned member, the missing Actual and Ready-CRM contributions are correctly treated as no contribution in the forecast equations; the EUR 42,550.00 Accrual therefore remains fully reconciled to the EUR 42,550.00 Base Full-Year Forecast.

Business Unit Financial QA Result

The final financial equations remain internally consistent under every governed Business Unit filter context. No Business Unit selection produces a residual variance, and the Unassigned member remains isolated rather than being allocated to a real Business Unit.

Inference – Business Unit Financial QA

The zero-variance result proves that the Business Unit slicer is not merely changing visual labels; it is propagating a coherent financial filter context through Actual Sales, Project Accrual and Ready CRM measures. The same algebra that reconciles the company total also reconciles each governed organizational slice.

17.54 Completed Final Financial Reconciliation Sign-Off

The final financial reconciliation sign-off is now complete. At the all-company level:

$$\text{EUR } 14,635,067.81 + \text{EUR } 6,616,025.78 = \text{EUR } 21,251,093.59$$

$$\text{EUR } 21,251,093.59 + \text{EUR } 807,900.00 = \text{EUR } 22,058,993.59$$

The corresponding DAX reconciliation measures return EUR 0.00. The same result was then confirmed under ExEd Programs BU, Qualification Programs BU, University Programs BU and Unassigned / Missing BU. The completed sign-off therefore covers both company-level reconciliation and Business Unit-level filter propagation.

Final Financial Reconciliation Result

The governed Base Full-Year Forecast and the Forecast Including Ready CRM now have an explicit zero-variance sign-off. The core financial logic is complete and no additional forecast calculation is required for the current interview case.

Inference – Final Financial Sign-Off

The forecasting solution has passed source reconciliation, dimensional relationship tests, interactive report filtering and final algebraic reconciliation. The remaining work is presentation-oriented rather than model-oriented: visual polish, default-state reset, checkpoint saving and clear communication of the business logic and unresolved governance exceptions.

17.55 Immediate Next Step – Final Presentation Polish and Interview Narrative

No new core semantic-model logic is required. The next work package is to prepare the finished interview version of the report:

1. exit **Edit interactions** mode and clear the Business Unit slicer so the Executive Dashboard opens on the full-company view;
2. align KPI cards and visuals, normalize spacing and font sizes, and ensure titles are fully readable;
3. keep the Forecast Governance table compact and preserve the distinction between Included, Held and Not Included values;
4. confirm that the Executive Dashboard opens with EUR 14.64M Actual, EUR 6.62M Accrual, EUR 21.25M Base Forecast and EUR 22.06M Forecast Including Ready CRM;
5. save a final Power BI checkpoint before the interview; and
6. prepare a short narrative covering source controls, forecast construction, unresolved CRM risk, Business Unit governance and the final management view.

Inference – Presentation Readiness

The analytical work is complete enough to shift attention from building to communicating. A clean default dashboard state and a concise explanation of what is included, held and excluded will make the technical controls understandable to Finance, data architecture and business stakeholders without adding any new forecast assumption.

18. Key Interview Message

The implementation does not treat the provided Excel files as simple reporting inputs.

It establishes a controlled forecasting process with raw-data staging, repeatable transformations, documented quality rules, grain validation, financial reconciliation and traceable analytical outputs.

The source Total is preserved as a control value instead of being silently deleted. Exceptional zero, negative and missing-classification records are retained transparently for Finance review rather than being hidden through automatic filtering.

The exact-duplicate control quantifies 8 groups, 20 involved source records and 12 excess records. It also estimates EUR 175,318.63 of potential overstatement without altering the source-aligned forecast before business validation.

The broader repeated-key control identifies 10 project-month groups containing 32 source records. It distinguishes four same-amount groups from six groups containing multiple amounts, demonstrating why repeated records must be classified before any removal decision is made.

The completed **RepeatedKeyStatus** classification further separates the results into 4 same-amount groups, 4 mixed groups and 2 groups containing only multiple distinct components. This preserves the financial meaning of the source while directing Finance review to the correct exceptions.

The CRM opportunity source has also been profiled across all 126 records and 14 columns. The profile confirms complete opportunity descriptions, three Business Unit values and reliable numeric value fields, while exposing missing probability and annual-allocation values that require explicit quality rules.

Explicit analytical data types have now been assigned to all 14 CRM columns. The conversion retained all 126 records and produced zero errors. This provides a technically stable basis for date analysis, probability calculations and financial validation while preserving missing business values for transparent quality classification.

The four CRM text columns were standardized using Trim and Clean, and all 14 source fields were assigned consistent analytical names. The subsequent **OpportunityQualityStatus** control classified every opportunity with zero errors and zero empty statuses. It identified 95 valid records, 26 records without annual allocation, 2 zero-value records, 2 records with missing probability and 1 record whose combined allocation exceeds 100%. No source row was deleted automatically; the findings remain visible for controlled business review.

An independent CRM weighted-value reconciliation now confirms 123 matches within a EUR 0.01 tolerance. Two records remain not testable because Probability is missing, while one material mismatch has been isolated with a source-minus- calculated variance of EUR -468,750.00. The source value has not been changed; the exception is preserved for targeted business validation.

The comparison between **OpportunityValue** and **EstimatedValue** has also been converted from a visual observation into an explicit control. All 126 CRM records return **Match** within the EUR 0.01 tolerance, with zero errors and zero empty classifications.

The current-year weighted amount is now independently recalculated from estimated value, probability and the explicit current-year allocation. The resulting **CurrentYearWeightedVariance** contains zero errors, and the final **CurrentYearWeightedCheck** is 100% valid with zero errors and zero empty classifications. The dedicated status-summary query confirms 85 **Match** records, 39 records that are not testable because the current-year allocation is missing, and 2 records that are not testable because Probability is missing. No current-year **Mismatch** is present among the 85 fully testable records.

A Power Query dependency-order issue was also resolved transparently: the current-year variance is now created before the final check that references it. No source value was changed to solve the technical issue.

The broader annual-allocation control now classifies all 126 CRM opportunities. Seventy records have both current-year and next-year allocations and sum to exactly 100%. Twenty-nine records have both allocations missing, 12 are missing only the current-year allocation and 11 are missing only the next-year allocation. These complete records sum to less than

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